



南昌大学
NANCHANG UNIVERSITY

第七届“计算成像技术与应用”研讨会

面向物理属性复用的生成式智能计算光学成像



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南昌大学人工智能工业研究院

2025.5.23-25 | 中国·上海

- 01 | 计算光学成像中的物理属性
- 02 | 复用技术 — 从信号处理到医学频域成像
- 03 | 复用技术 — 从医学到计算光学成像



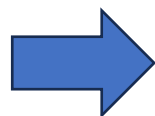
计算光学成像中的物理属性



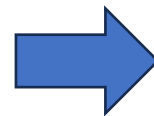
- 基于几何光学理论的光学工业设计方法简单易用
- 但其线性近似方法已**达到理论极限**，提升空间有限
- 数字信号处理与计算机技术的发展，为**成像链路中引入信息处理**提供可能

信息驱动的计算成像技术

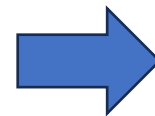
打破物像共轭
一对一 ➡ 一对多



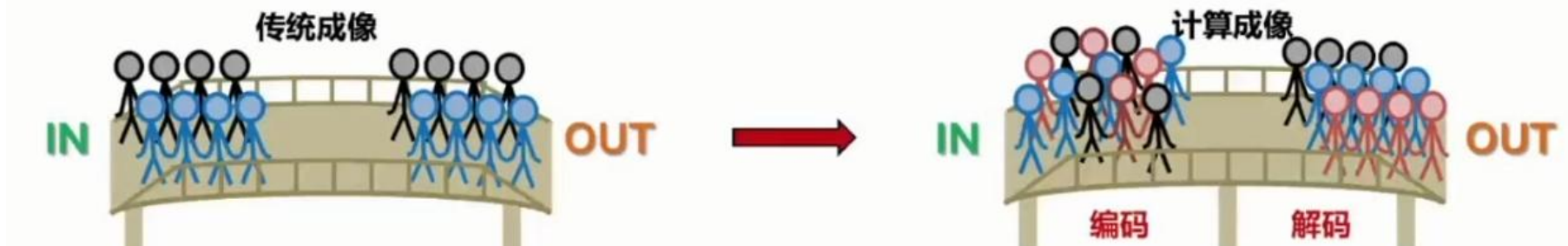
信息混叠
不完美编码数据



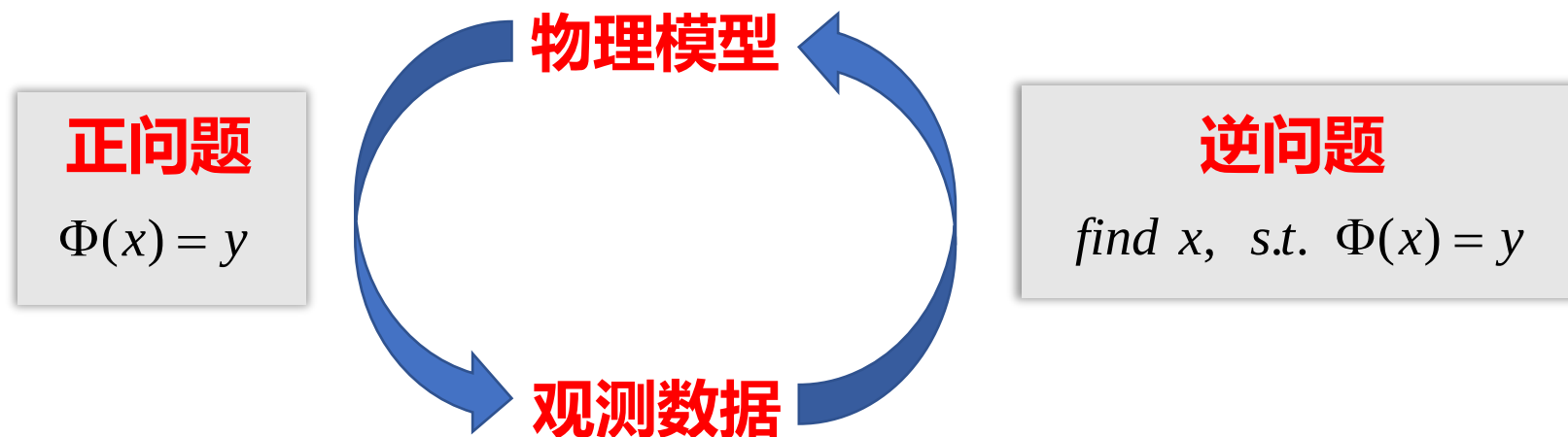
计算重建



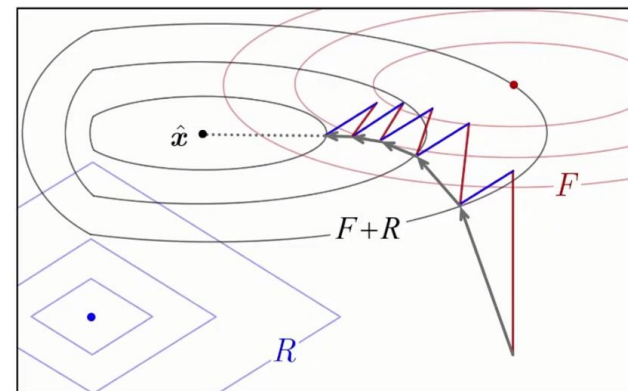
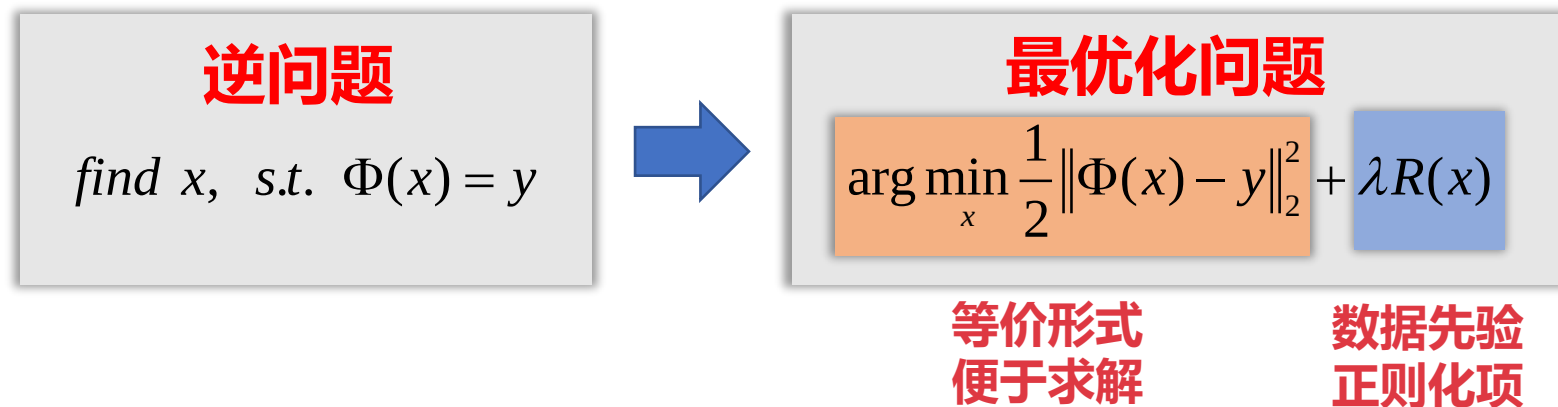
信息恢复
完美图像

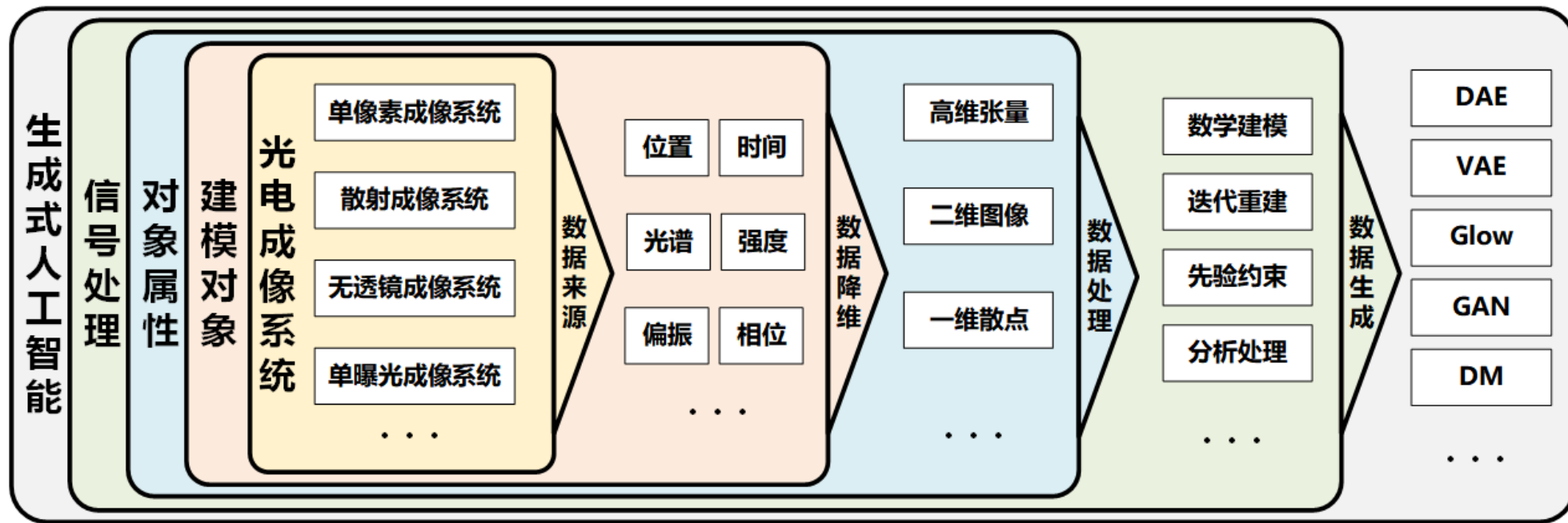


- 基于物理结构的正向模型，建立信息重建逆问题



- 转变为便于求解的优化问题





- 利用扩散模型估计光场多变量分布的特点
- 通过在空域、频域、多模态域、时域上不同变量的组合
- 通过高维及相关这两个属性，更好地对拟重建对象进行信息挖掘
- 构建新的人工变量，与对应的扩散模型，达到利用其先验信息的目的

生成模型-----从给定分布 $P_{data}(x)$ 取样的若干样本构成训练集

模型学习到一个模拟这一分布的概率分布 $P(x)$

01

Denoising autoencoding(DAE)

02

Variational Autoencoders (VAE)

03

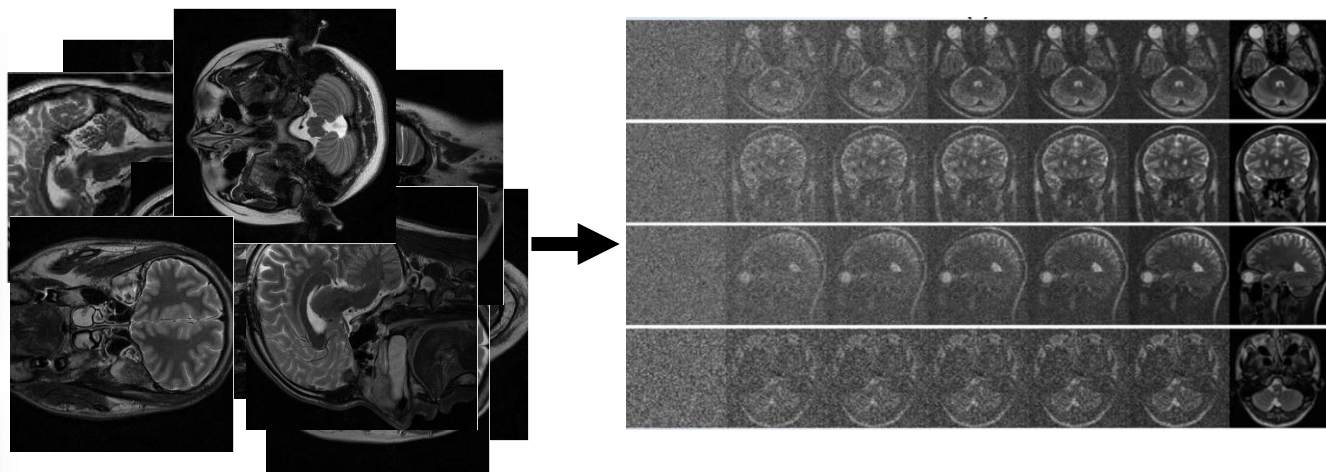
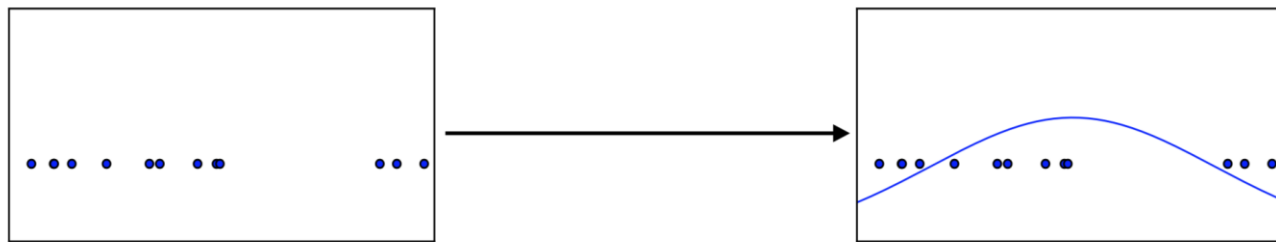
Generative Adversarial Network
(GAN)

04

PixelCNN

05

Generative Flow (Glow)

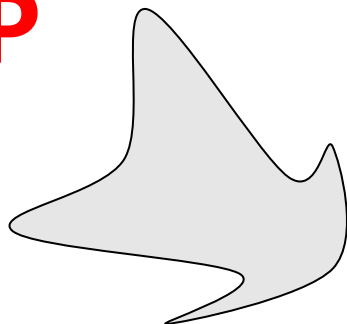


Q. Liu, Q. Yang, H. Cheng, S. Wang, M. Zhang, D. Liang, Highly undersampled magnetic resonance imaging reconstruction using autoencoding priors, *Magn. Reson. Med.*, vol. 83, no. 1, pp. 322-336, 2020.

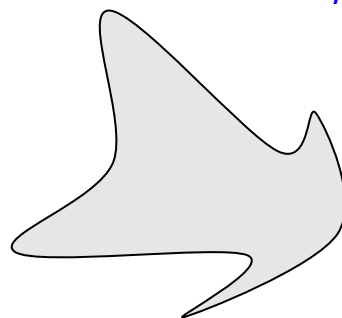
DAE

S. Wang, J. Lv, Z. He, D. Liang, Y. Chen, M. Zhang, Q. Liu, Denoising auto-encoding priors in undecimated wavelet domain for MR image reconstruction, *Neurocomputing*, vol.437, pp.325-338, 2021.

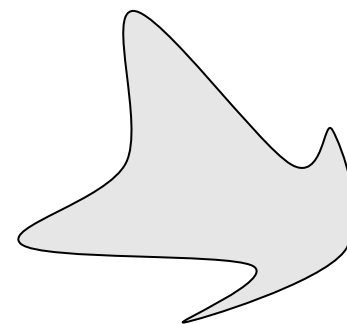
DAEP



DMSP



DSM



M. Zhang, M. Li, J. Zhou, Y. Zhu, S. Wang, D. Liang, Y. Chen, Q. Liu. High-dimensional embedding network derived prior for compressive sensing MRI reconstruction, *Med. Image Anal.*, vol. 64, 101717, 2020.

DMSP

C. Quan, J. Zhou, Y. Zhu, Y. Chen, S. Wang, D. Liang, Q. Liu, Homotopic gradients of generative density priors for MR image reconstruction, *IEEE Trans. Med. Imag.*, 2021.

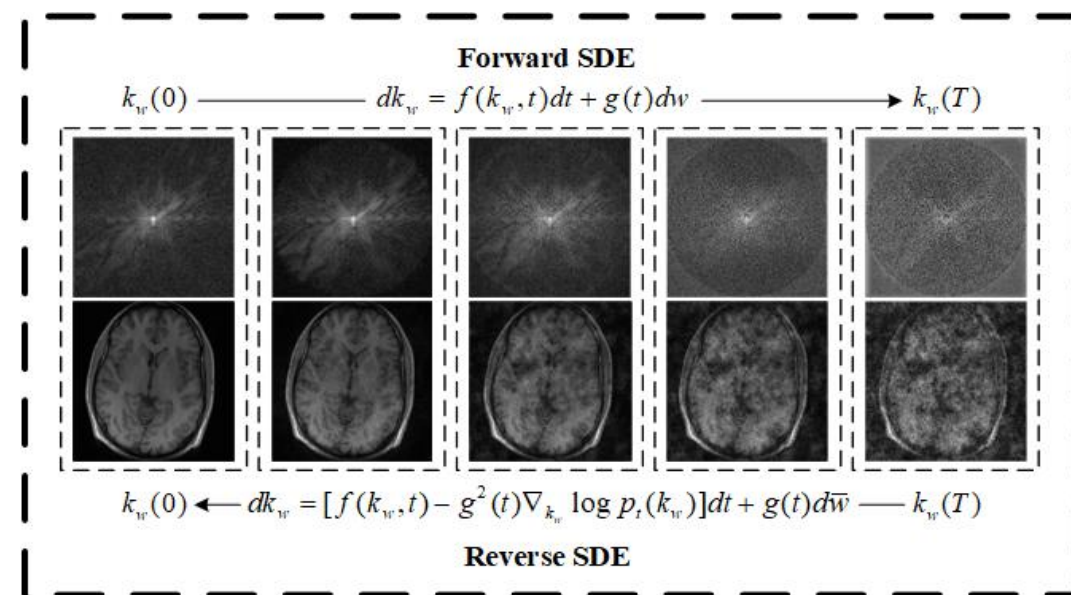
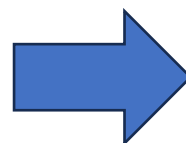
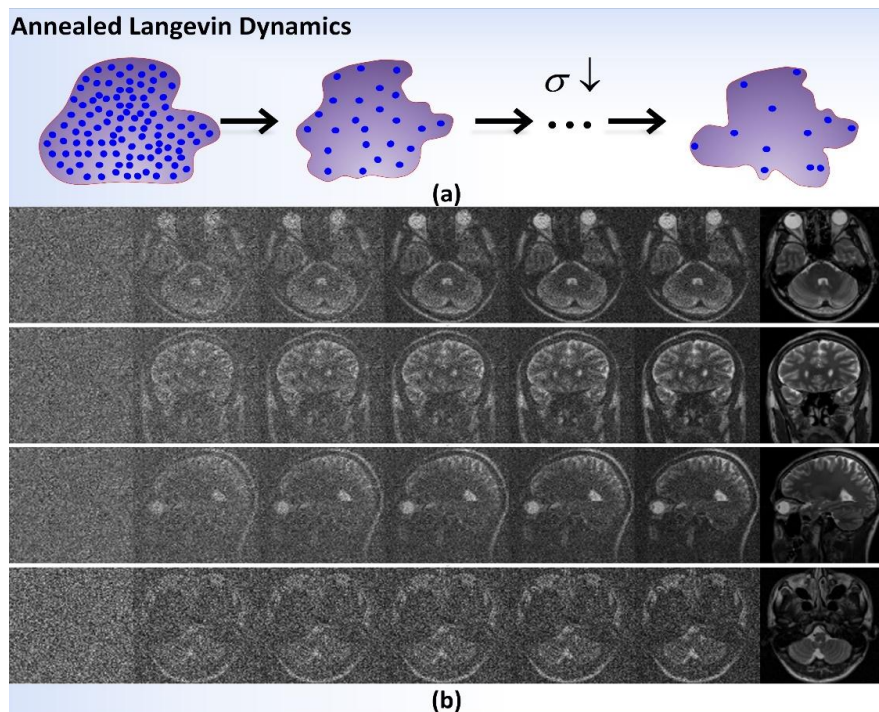
DSM

Generative model

复用技术 — 从信号处理到医学频域成像



Generative model in image domain

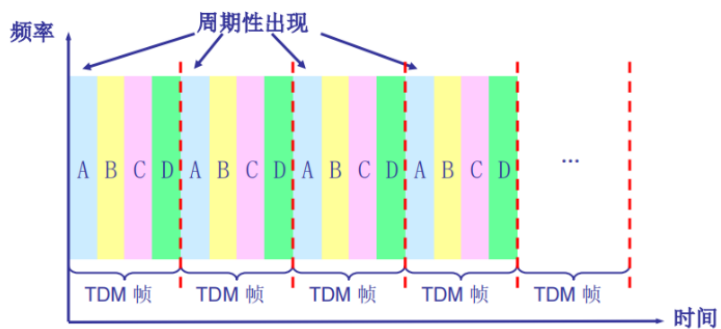


Generative model in k-space

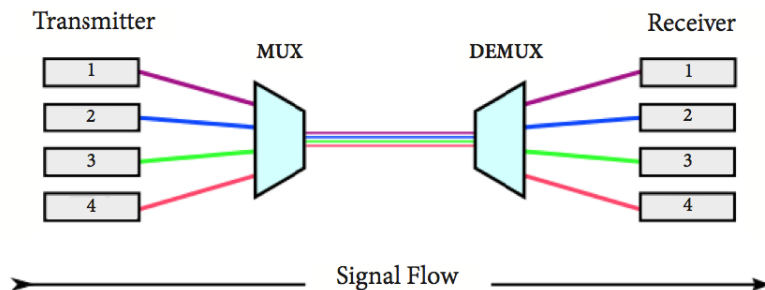
Z. Tu, D. Liu, X. Wang, C. Jiang, M. Zhang, Q. Liu, D. Liang, WKGM: Weight-K-space generative model for parallel imaging reconstruction, *NMR in Biomedicine*, e5005, 2023.

Lab. Imag. & Visual rEpresentation

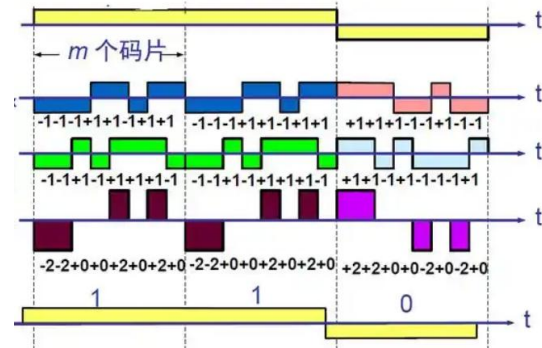
复用技术：通过一条通信线路上传输多路信号，从而提升线路利用率。



时分复用

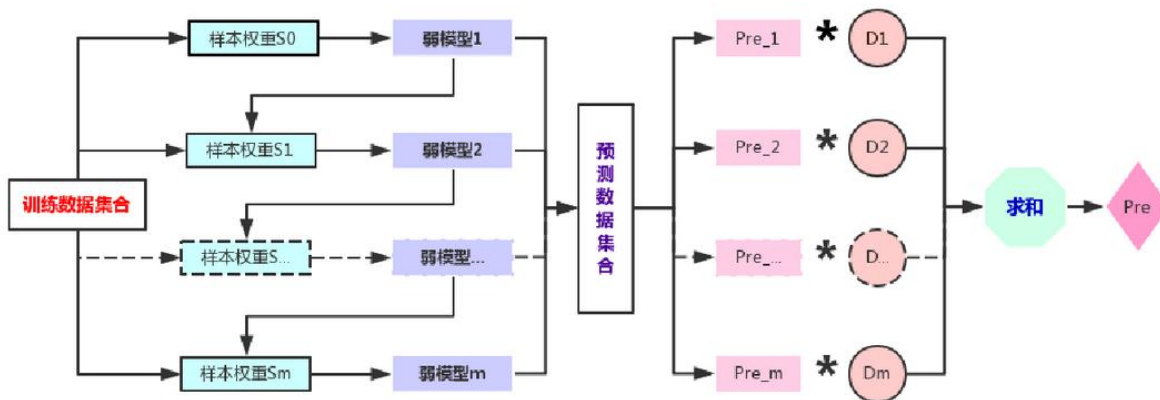


频分复用



码分复用

Boosting算法： 一组 “弱学习者” 的集合能否生成一个 “强学习者” ？



Multi-channel learning scheme (self-copy)

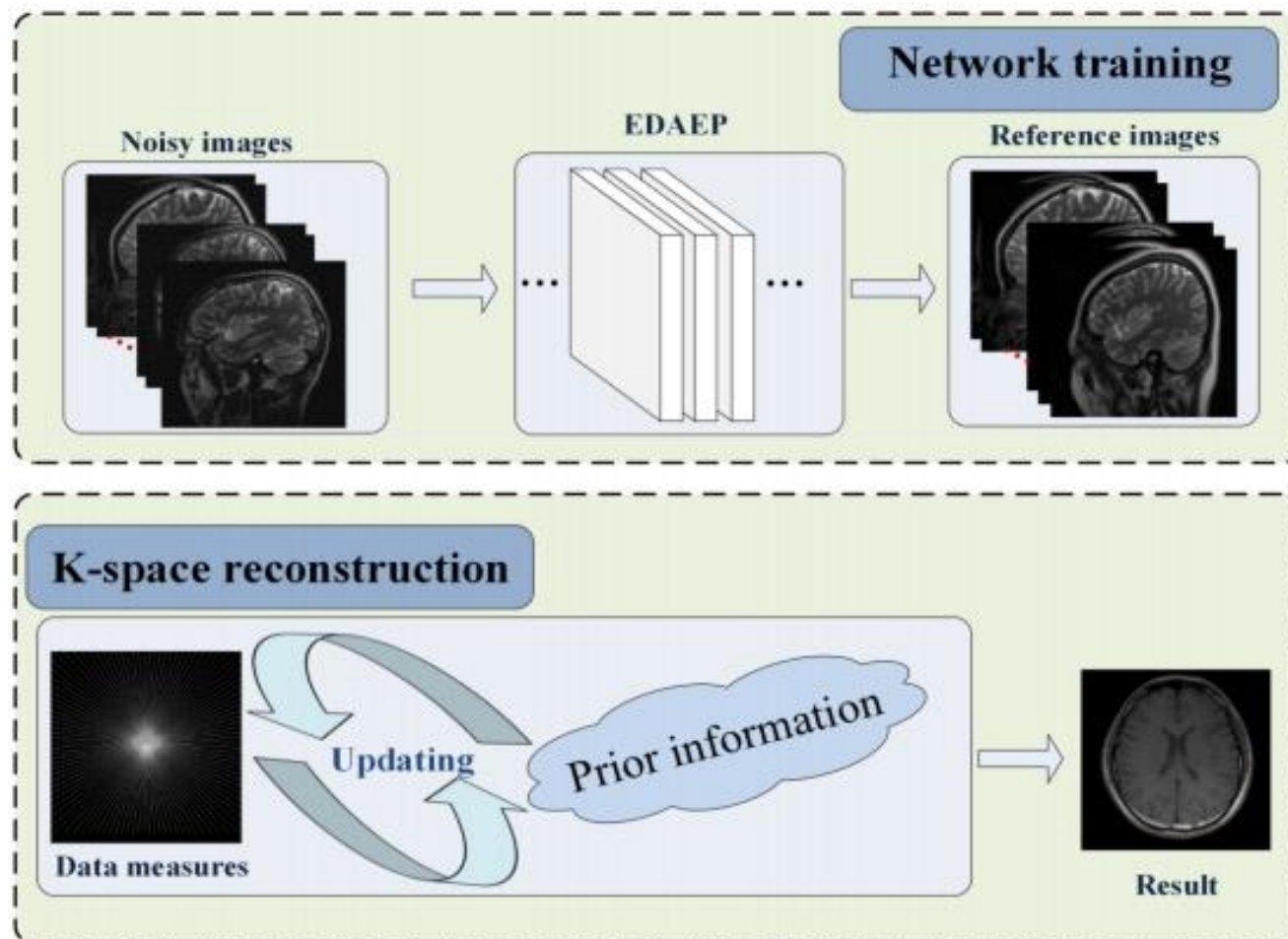
EDAEPRec

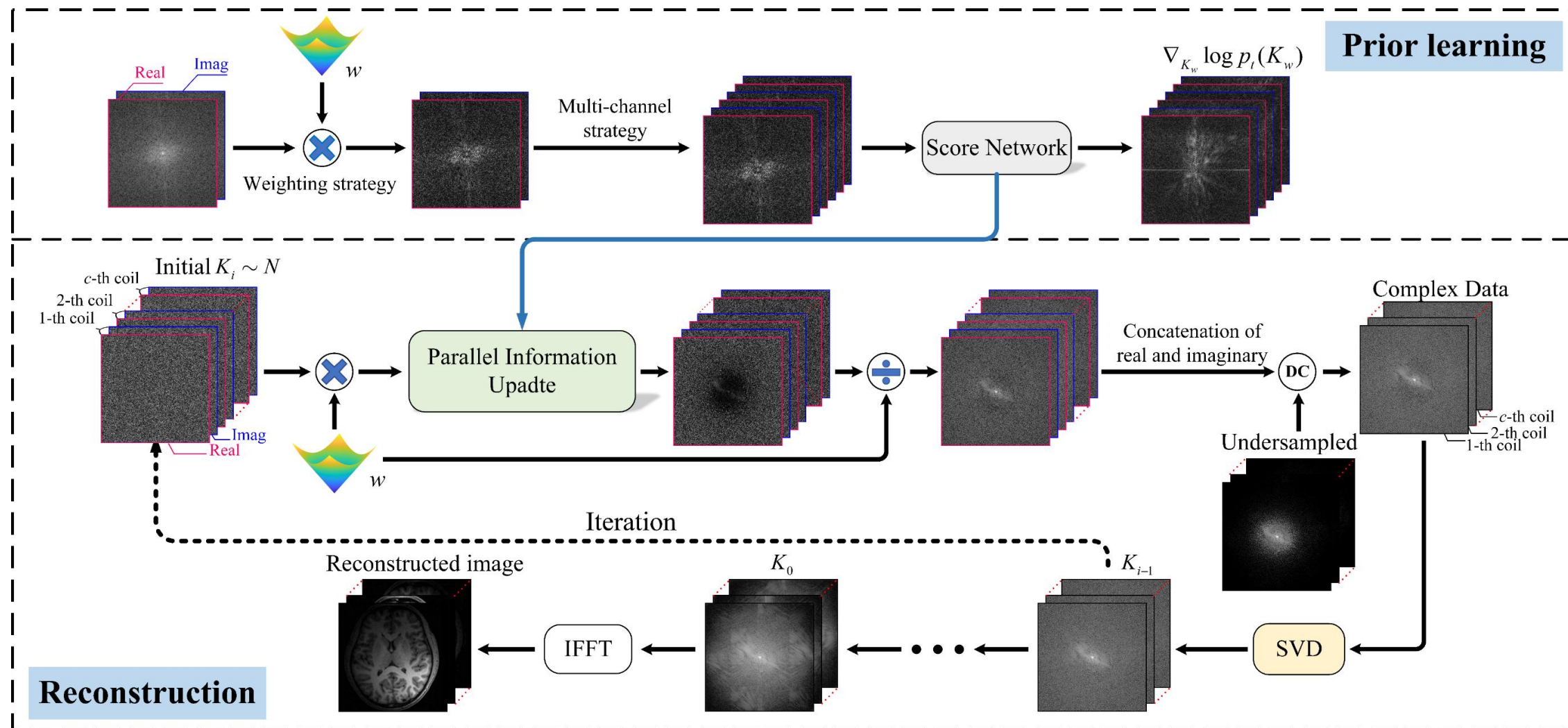
Training stage:

$$\{X = [x, x, x]\}$$

Testing stage:

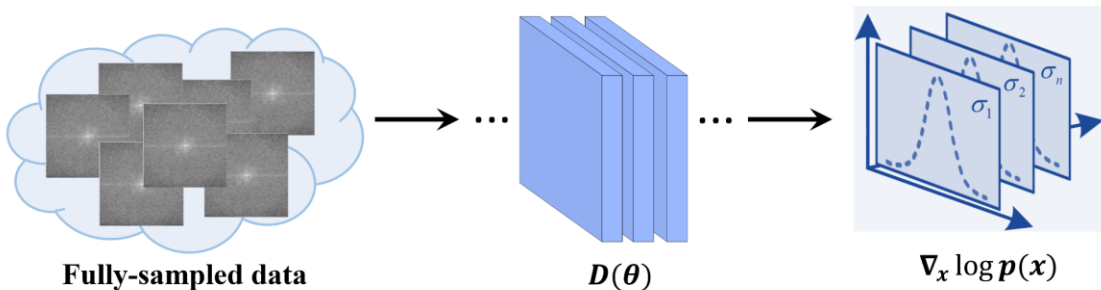
$$X^k = [x^k, x^k, x^k]$$





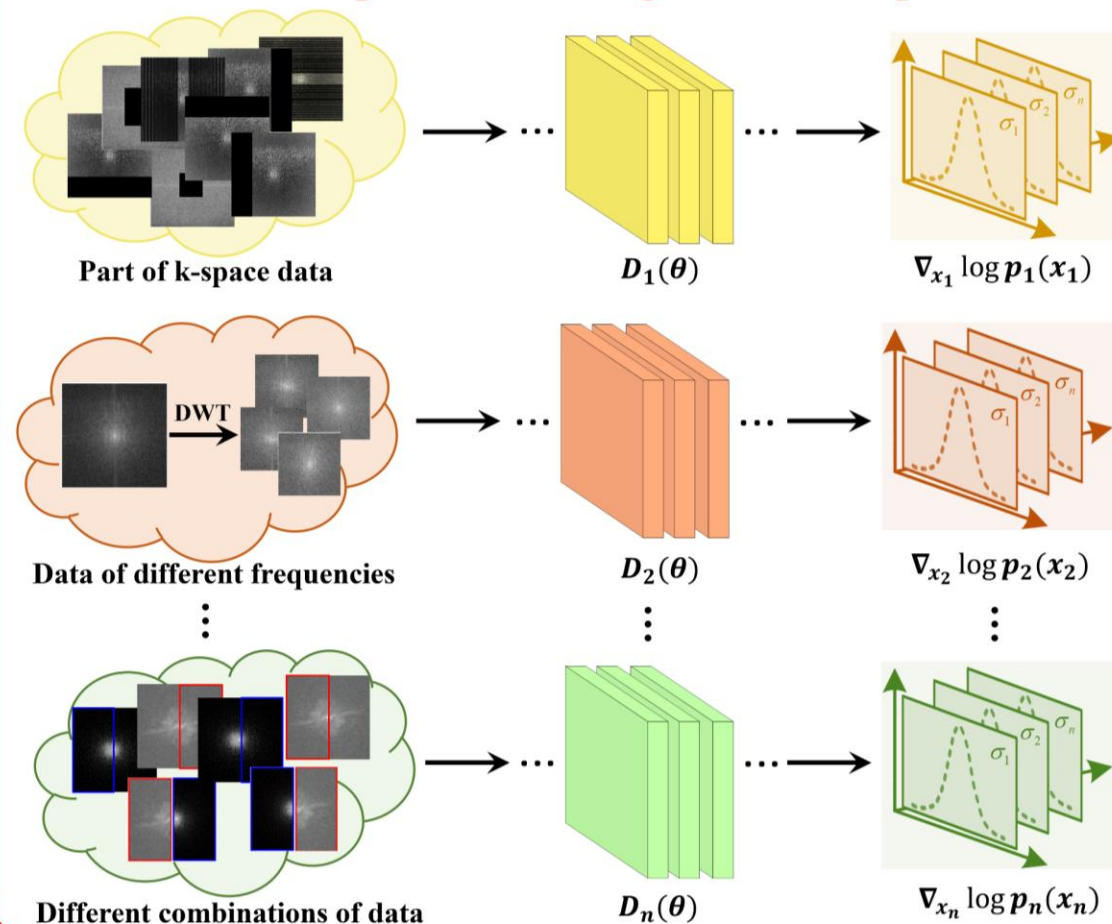
- 借鉴编码掩膜复用思想
- 单扩散模型 $\rightarrow$ 多掩膜扩散模型

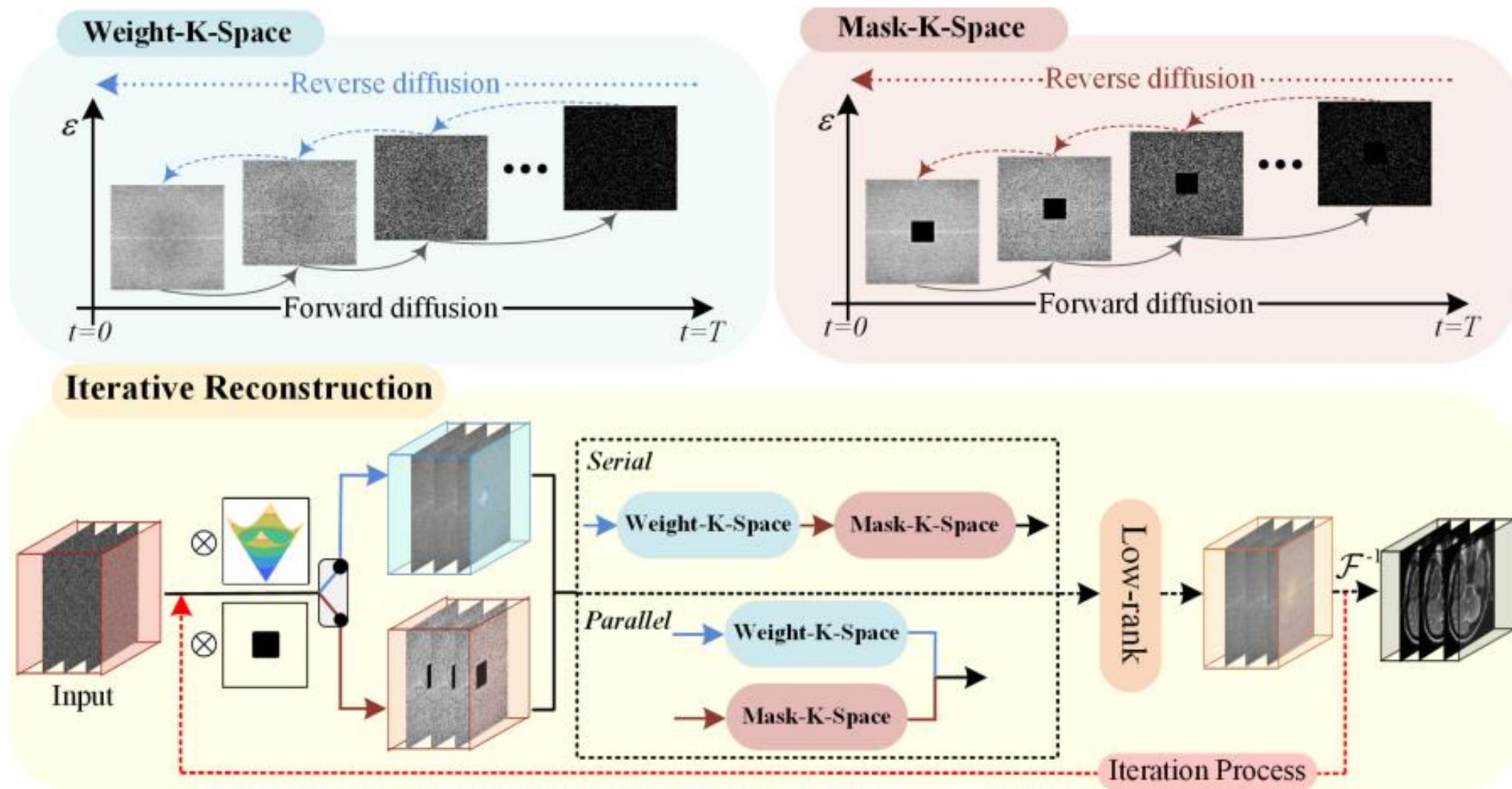
Train one model using fully-sampled k-space data



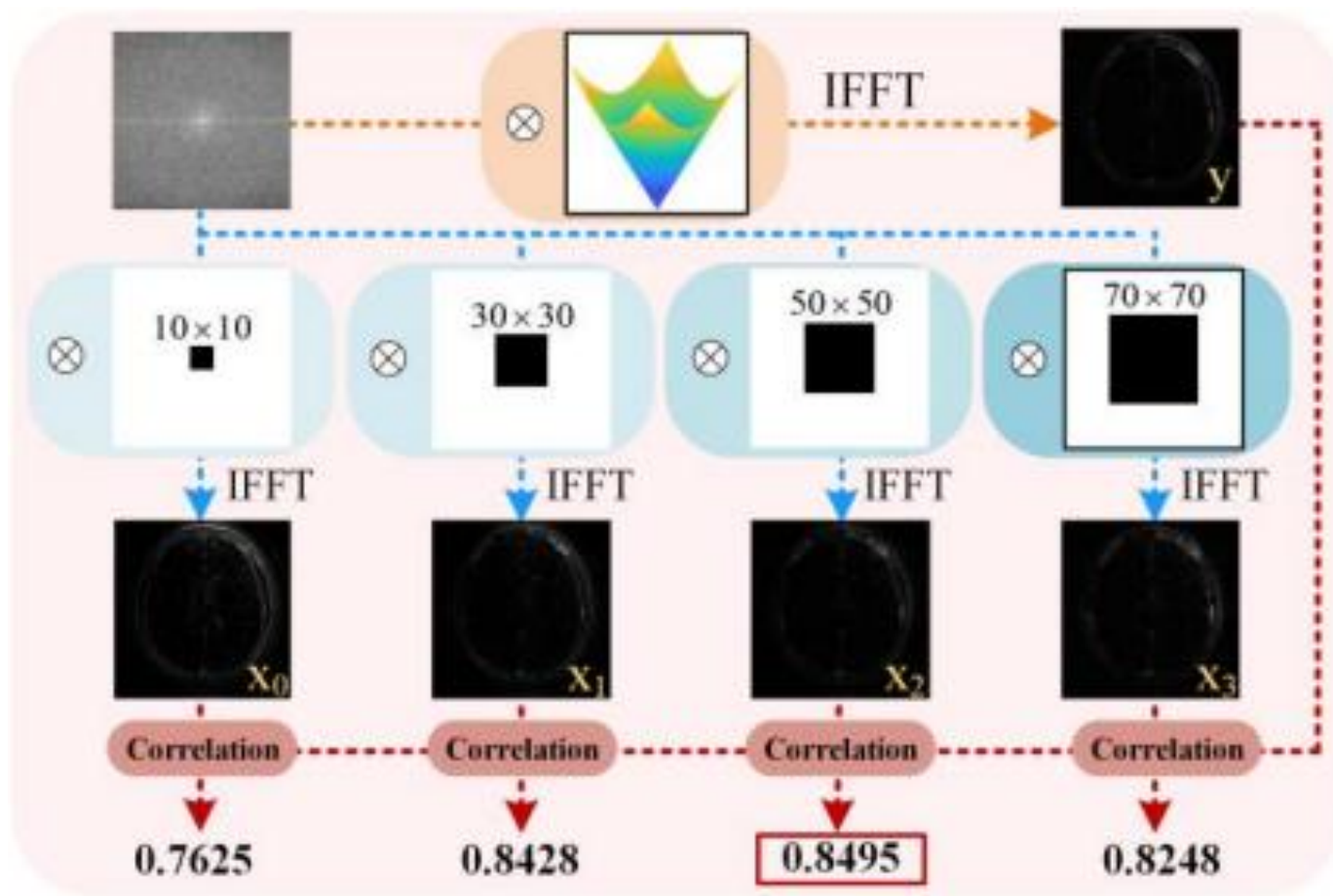
- 信息利用维度：
空间域、频域、小波变换域、多模态复用域

Train multiple models using different k-space data

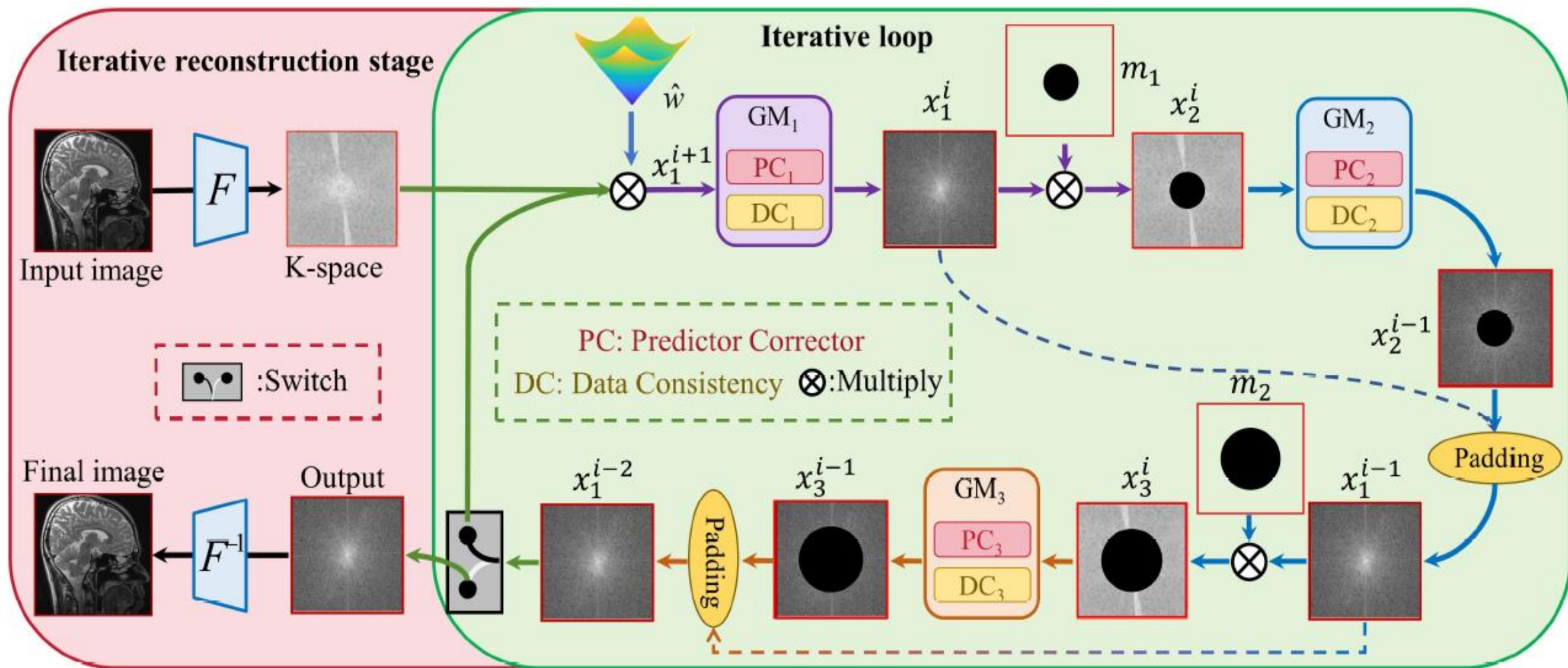


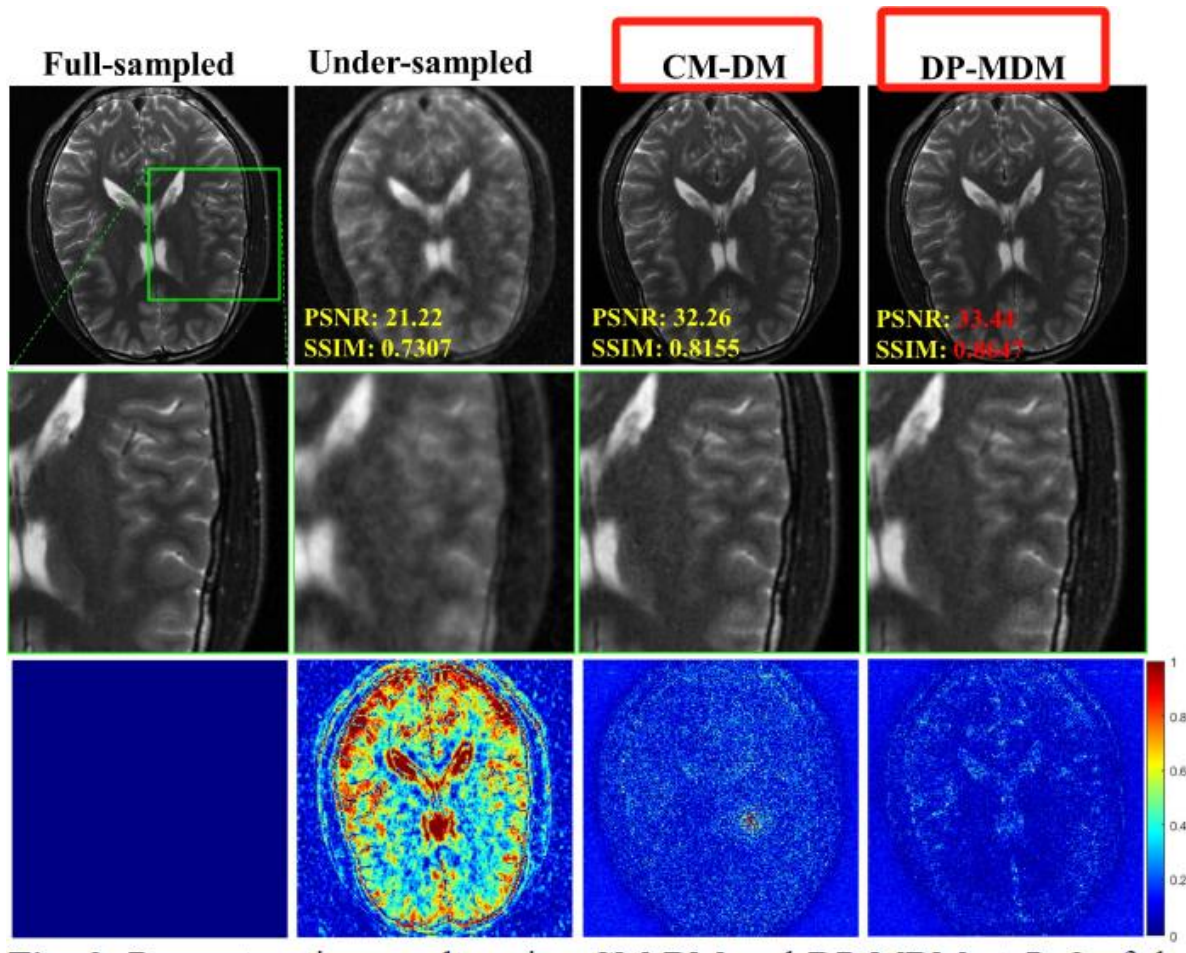


Guan Y, Yu C, Cui Z, et al. Correlated and multi-frequency diffusion modeling for highly under-sampled MRI reconstruction[J]. IEEE Transactions on Medical Imaging, 2024, 43(10): 3490-3502.



Guan Y, Yu C, Cui Z, et al. Correlated and multi-frequency diffusion modeling for highly under-sampled MRI reconstruction[J]. IEEE Transactions on Medical Imaging, 2024, 43(10): 3490-3502.





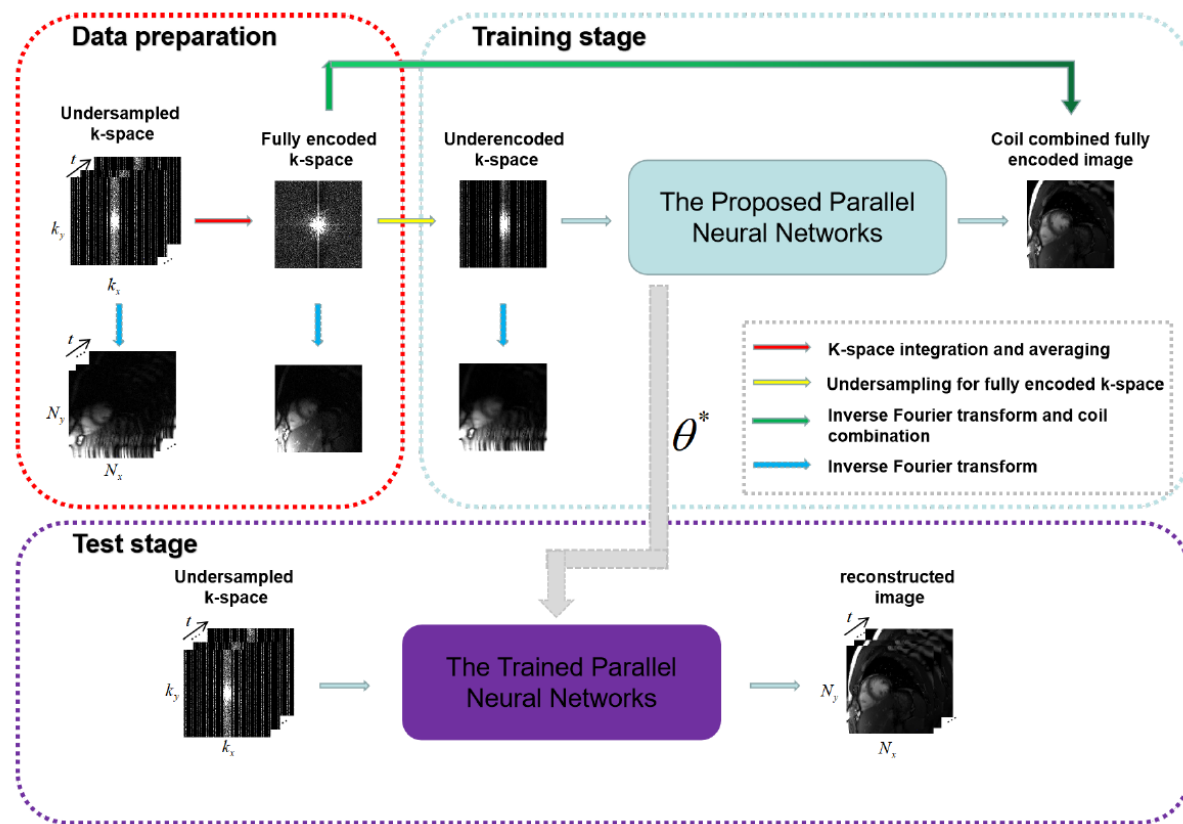
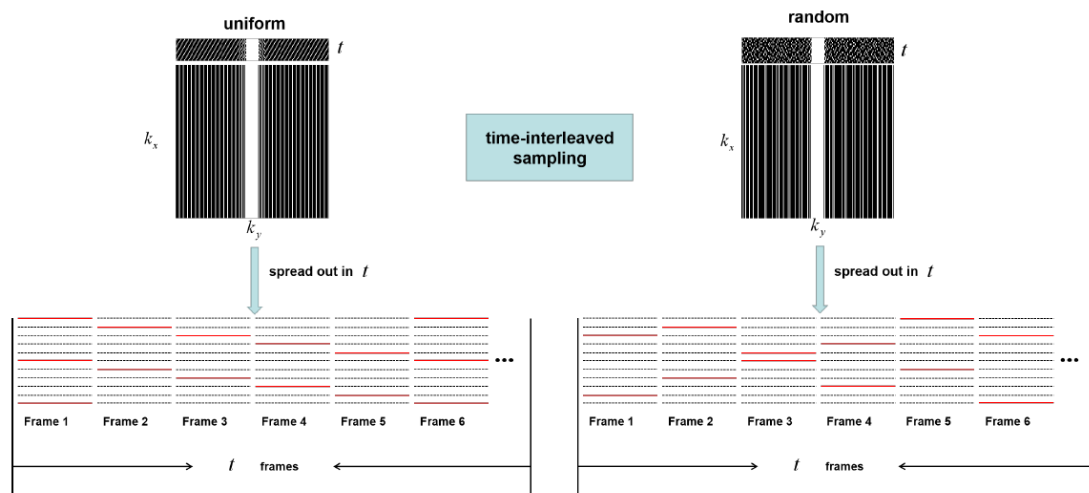
<i>Fast-MRI</i>	1-GM	2-GMs	3-GMs (DP-MDM)	4-GMs
Poisson $R=8$	29.59/0.7507	35.18/0.8990	38.05/0.9094	38.06/0.9104
Gap [*]	-----	5.59/0.1483	2.87/0.0104	0.01/0.0010
Poisson $R=10$	29.16/0.7449	35.18/0.8995	37.62/0.9001	37.67/0.9033
Gap [*]	-----	6.02/0.1546	2.44/0.0006	0.05/0.0032

*Gap represents the difference between the corresponding indicator values of the current column and the subsequent column.

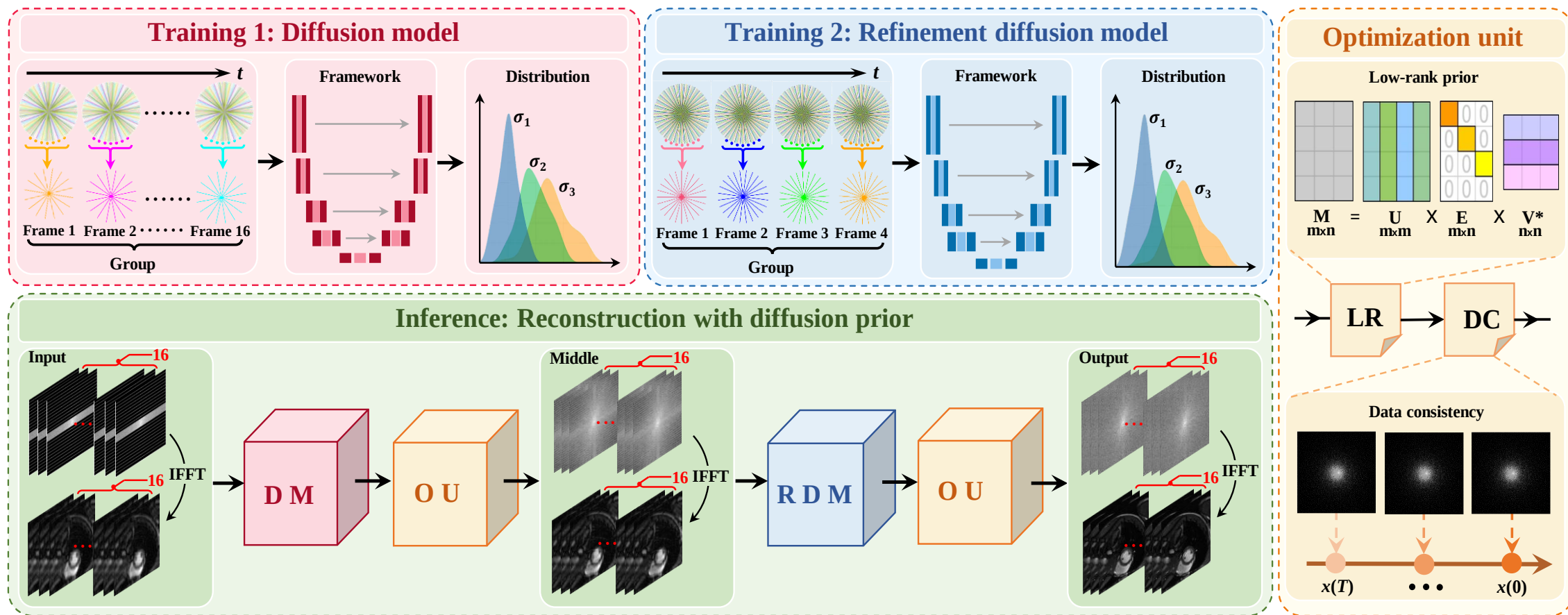
结果表明：
不同虚拟掩码以及不同模型个数对于
保留MR图像细节都有影响

- 聚焦时间维度的冗余信息，借鉴自监督工作的思想
- 利用多帧图像之间相似性的特点，自定义划分子集训练多个扩散模型

多帧图像的组合利用：

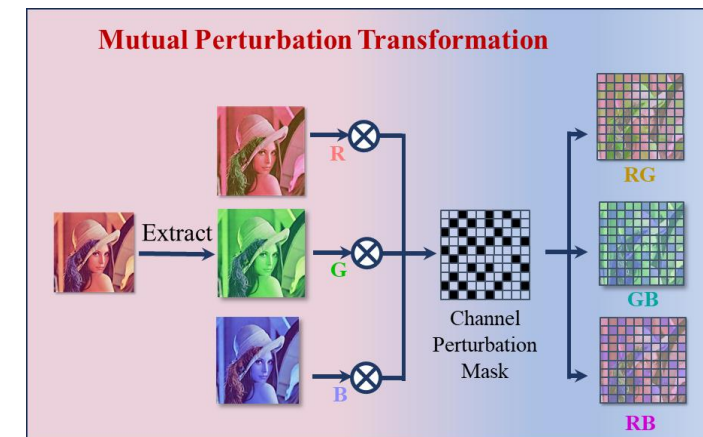
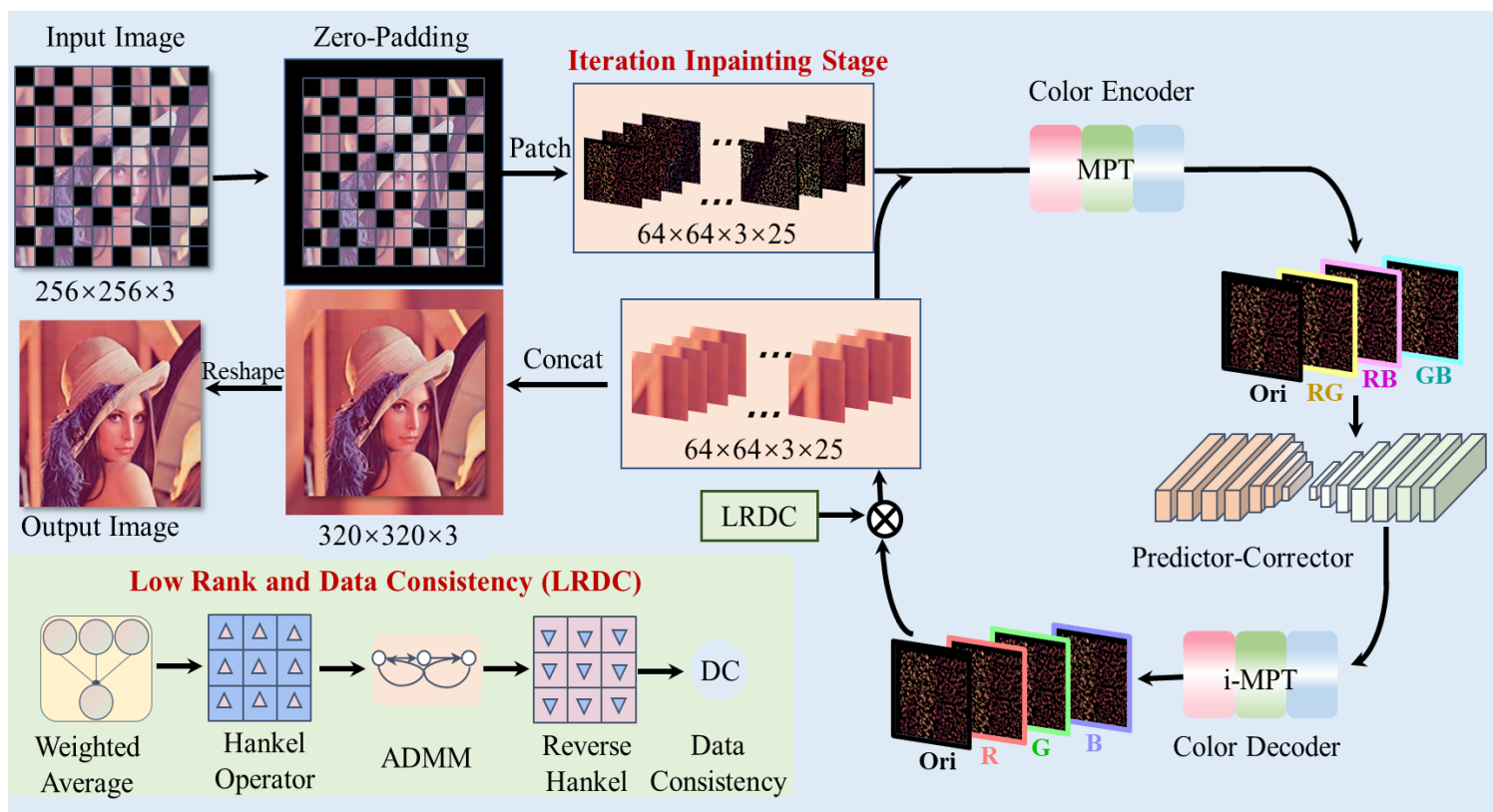


- 动态MRI成像在时间维度存在冗余信息量，例如多帧图像存在很多重复信息

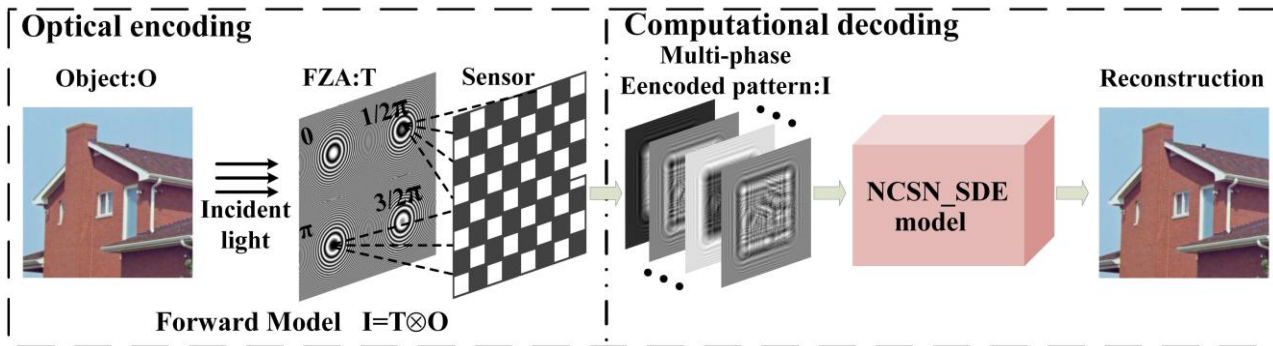


复用技术 — 从医学成像到光学成像





- 对图像的颜色编码进行了重新排列和调整
- 从而使得模型在训练过程中能够接触到更广泛和更复杂的编码模式

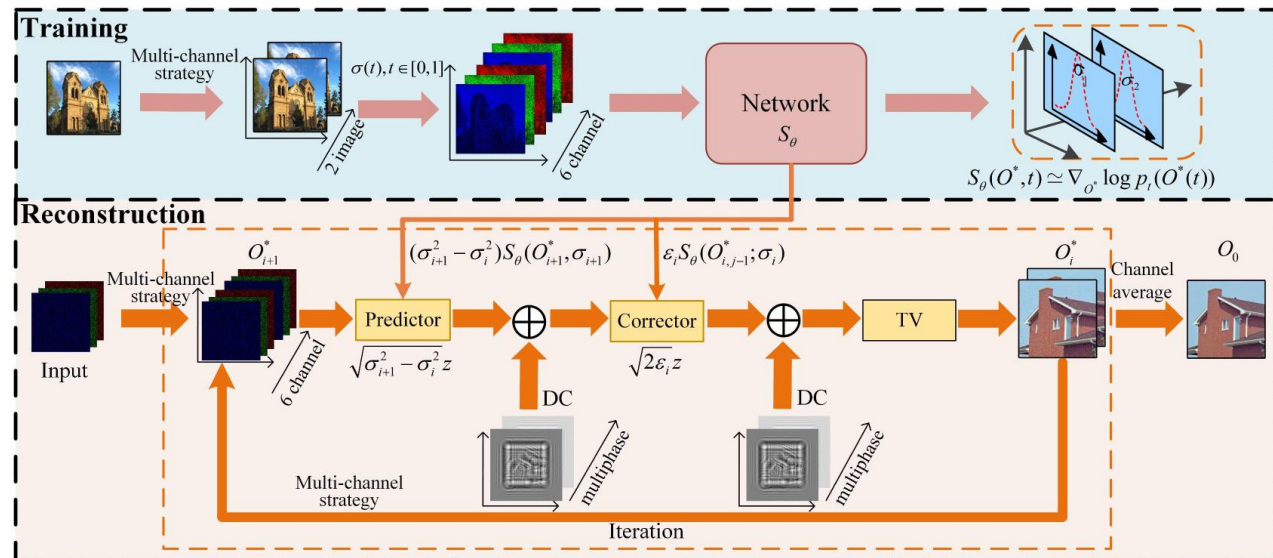


多相位编码:

- 用单帧数据获取多相位编码信息

先验学习:

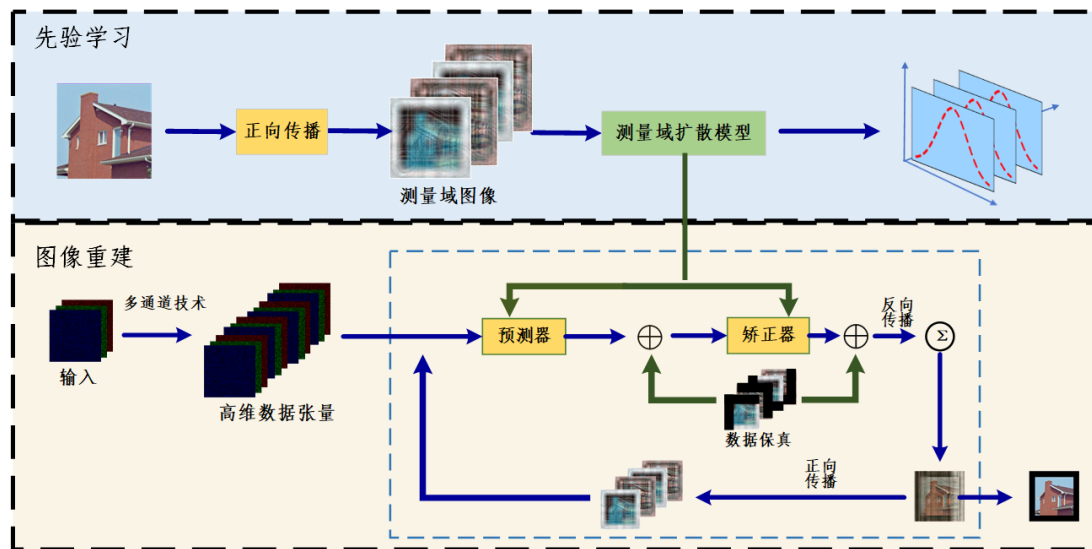
- 多通道技术——拓宽网络的学习通道
- 多色彩通道高维图像下的先验信息表示



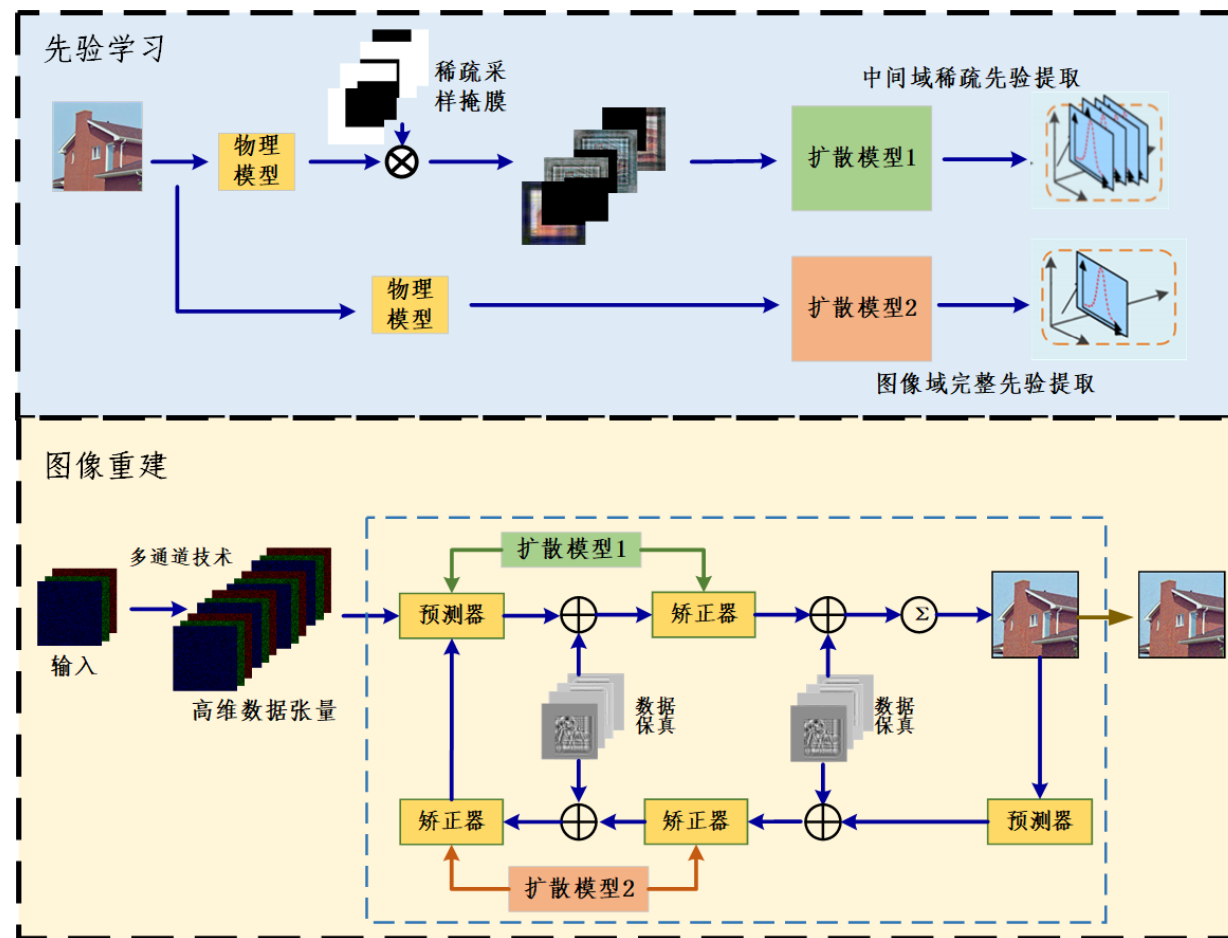
图像重建:

- 利用数据分布先验信息，约束基于物理模型的迭代重建过程
- 提高图像重建的质量

- 多相位编码不可避免带来**成像视场的降低**
- 使用四相位编码，视场将缩小至1/2
- 发展多相位互补稀疏采样无透镜成像

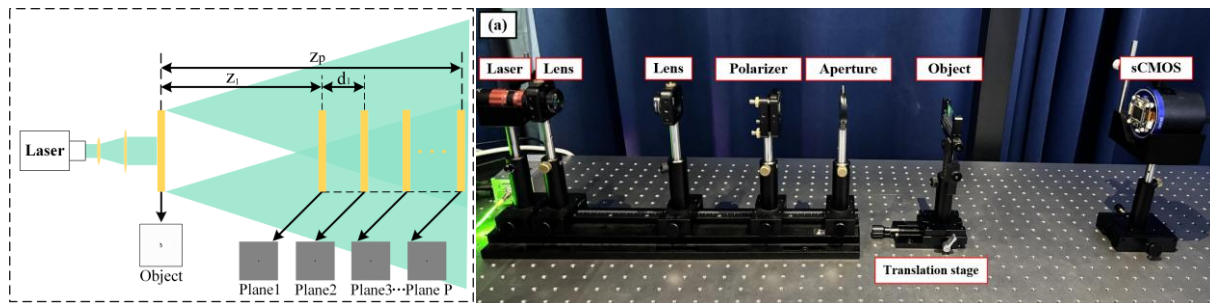


测量域多相位欠采样编码数据生成

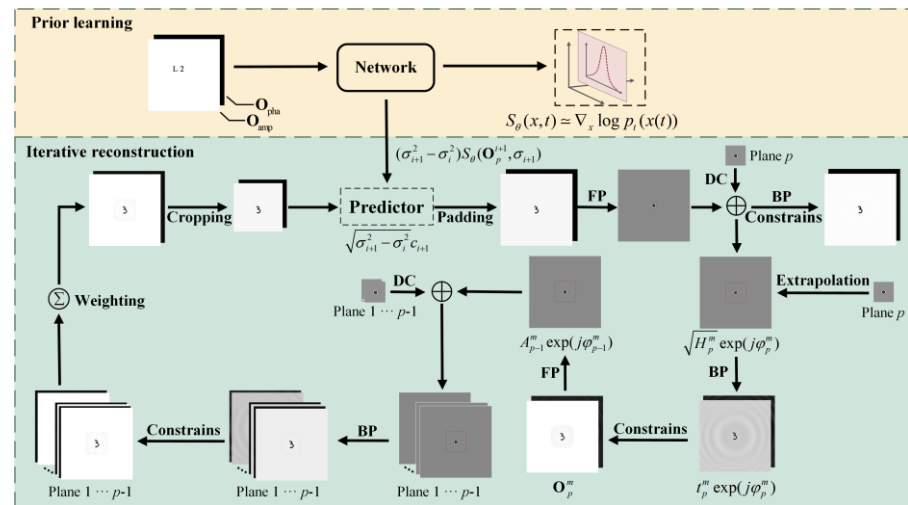
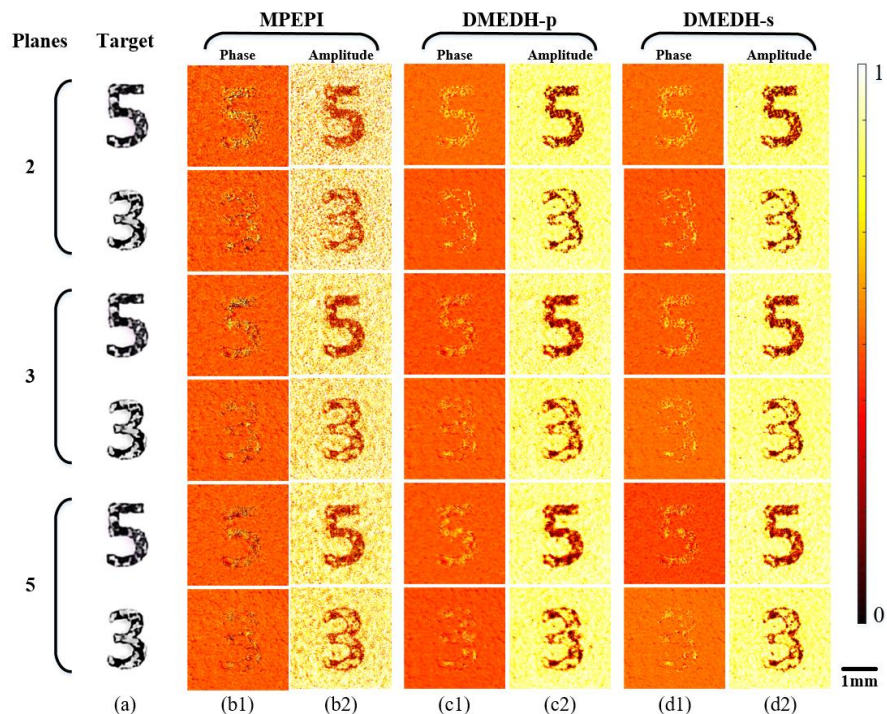


中间变换域-多相位编码-多掩膜复用扩散模型重建

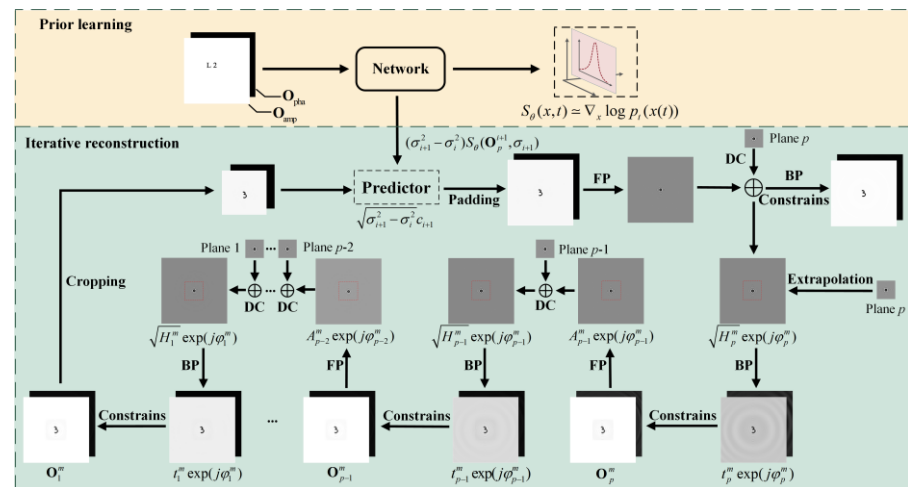
多距离编码复用的数字全息成像



多距离编码数字全息成像



并行加权信息利用策略

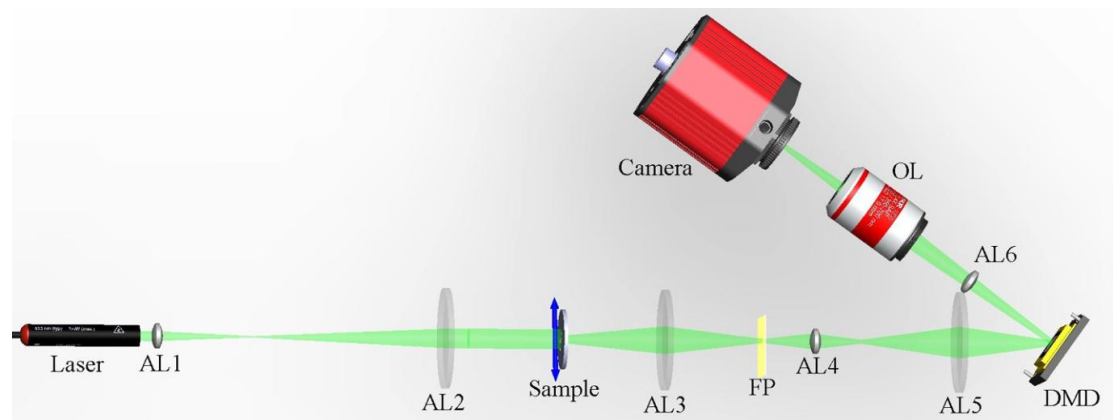


串行传输信息利用策略

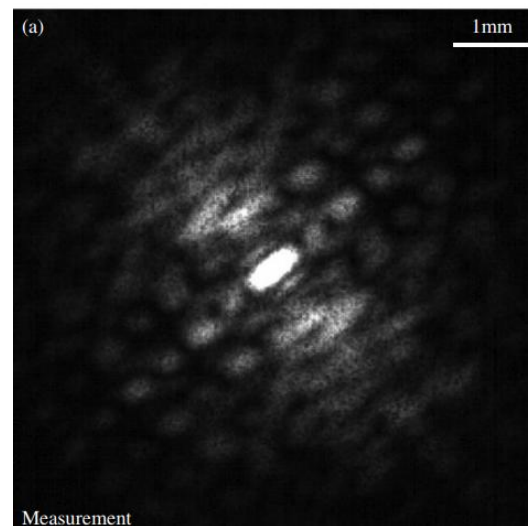
◆ 单曝光时间压缩相干衍射成像

- 西湖大学袁鑫教授提出TC-CDI技术^[1]
- 在频域采集目标数据
- 利用DMD压缩多帧频谱数据至一帧图像
- 实现超过相机帧率的相干衍射重建

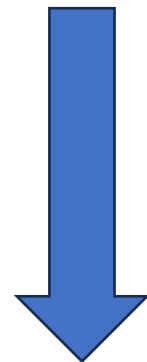
◆ 实验系统原理图



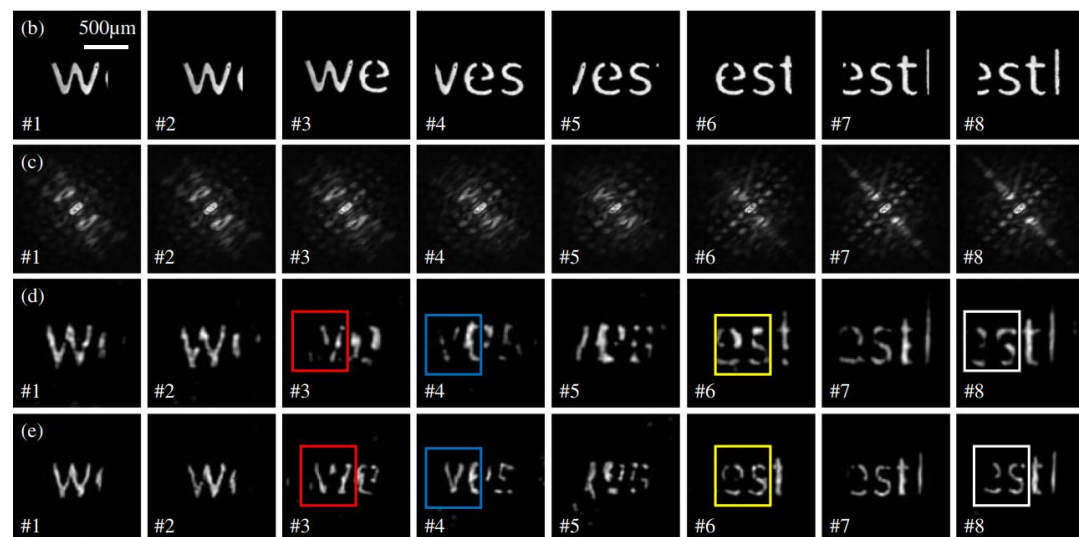
[1] Z. Chen, S. Zheng, Z. Tong, et al., “Physics-driven deep learning enables temporal compressive coherent diffraction imaging,” *Optica* 9(6), 677–680 (2022).



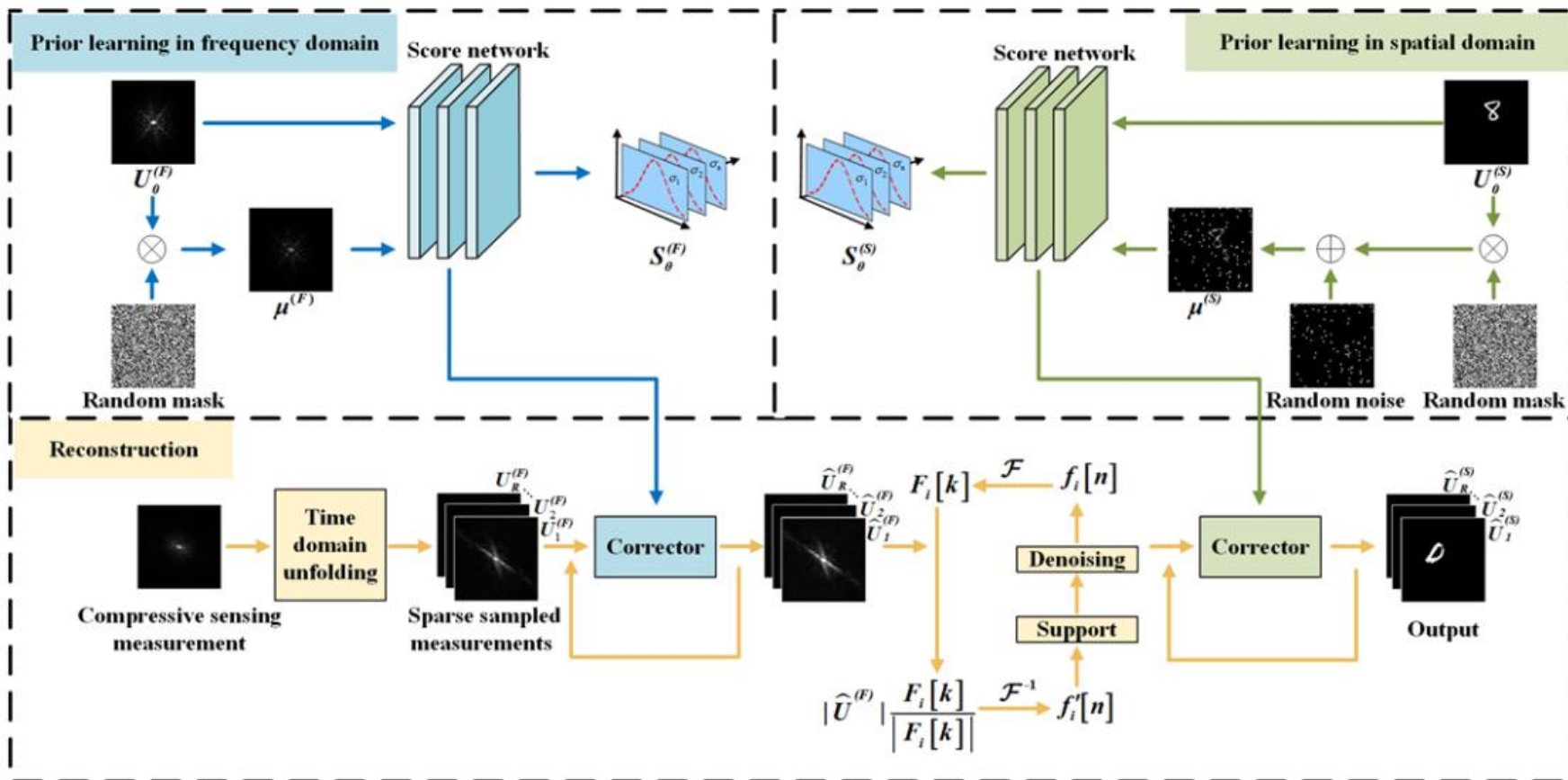
单帧频谱测量数据



多帧空域恢复数据



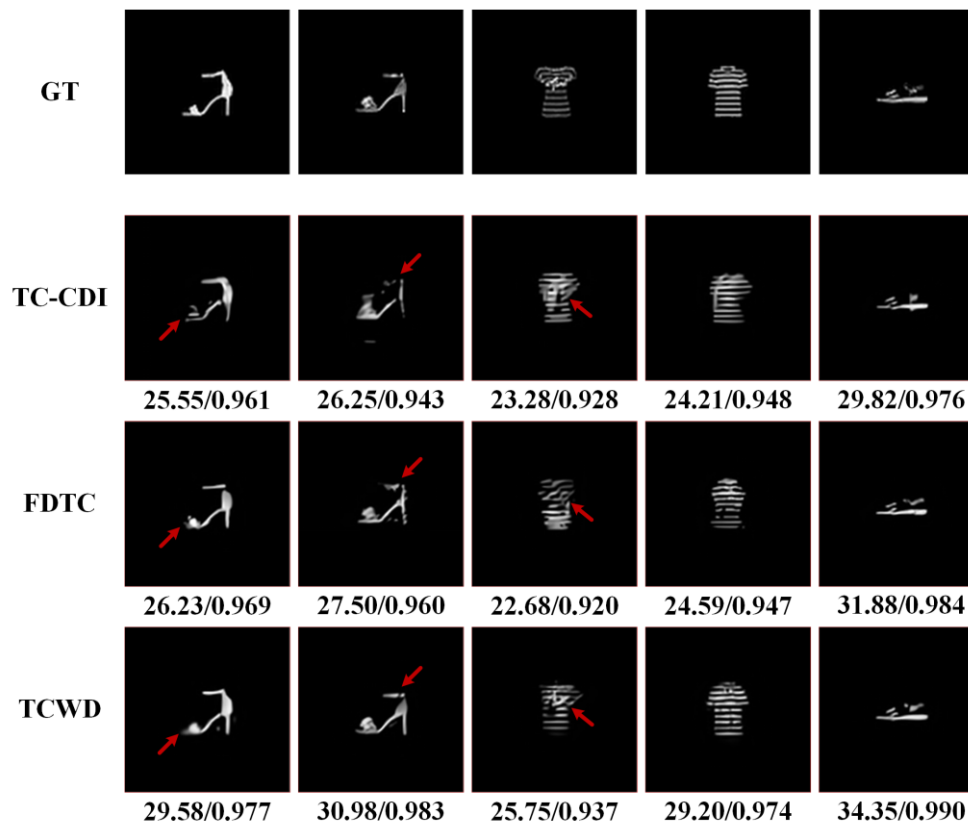
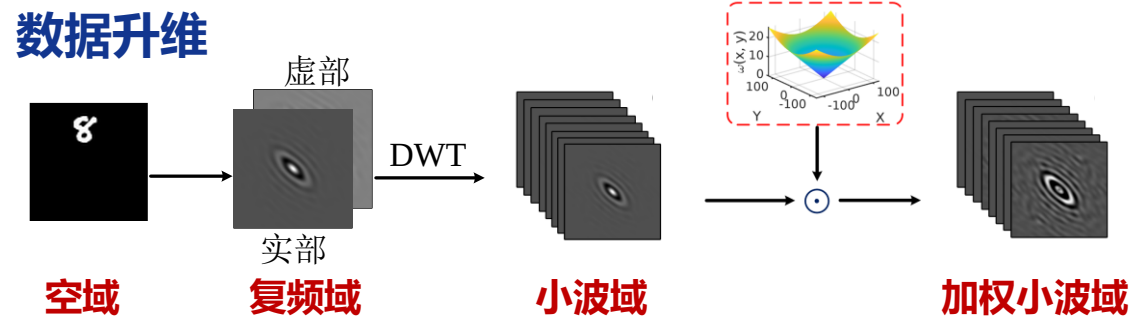
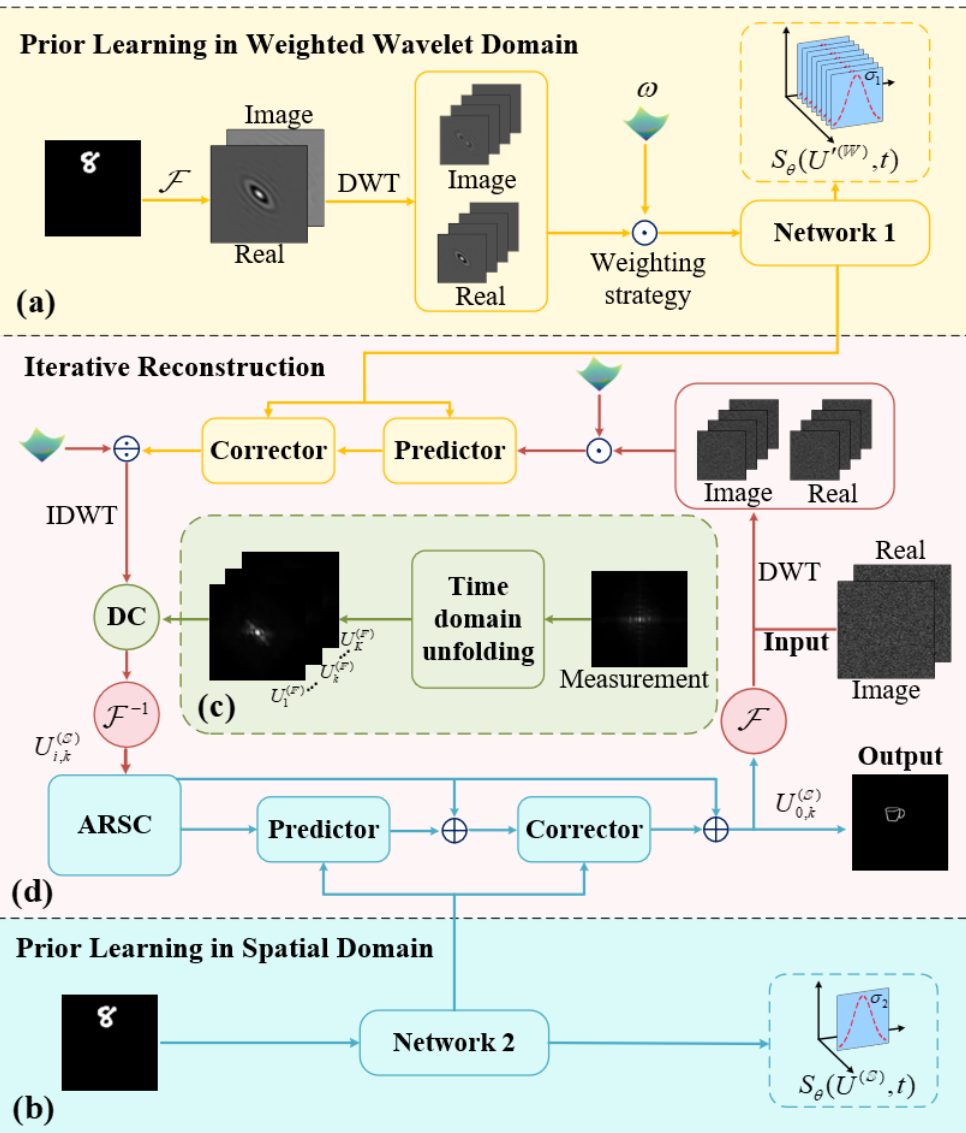
频域先验
信息学习

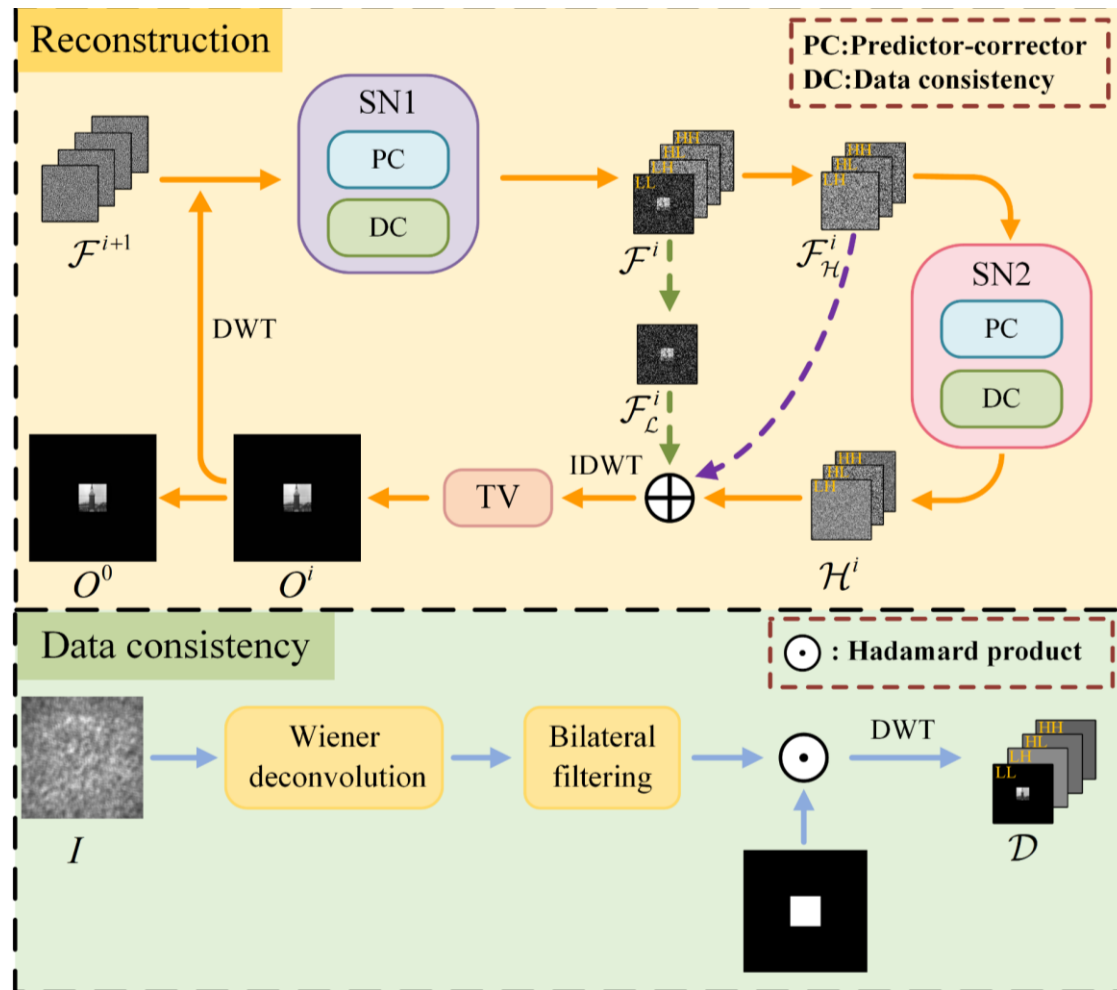
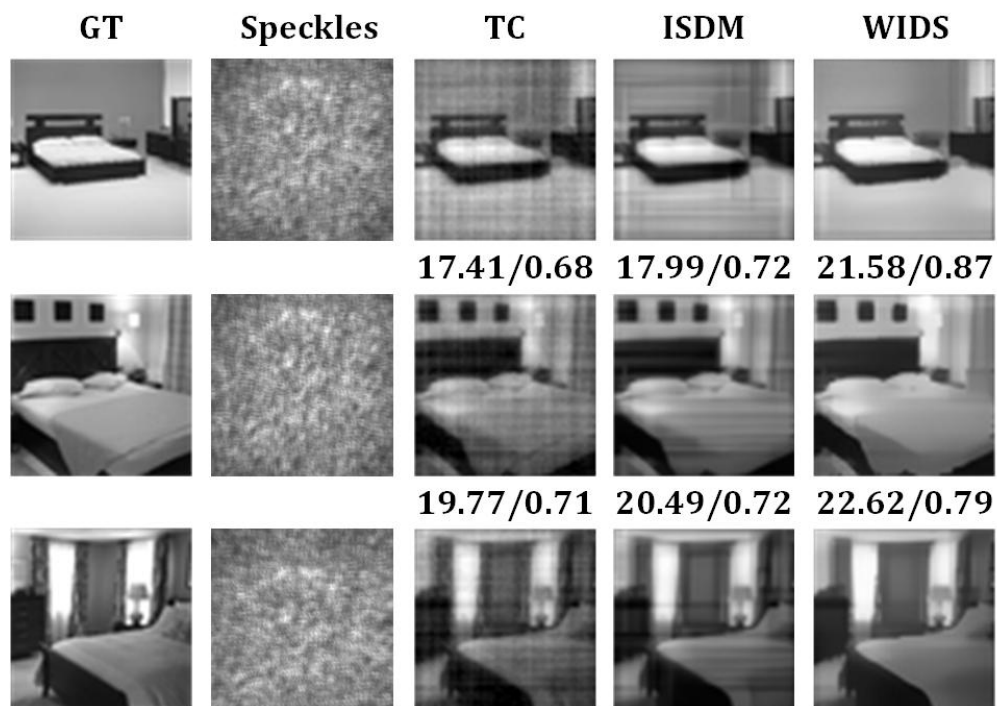
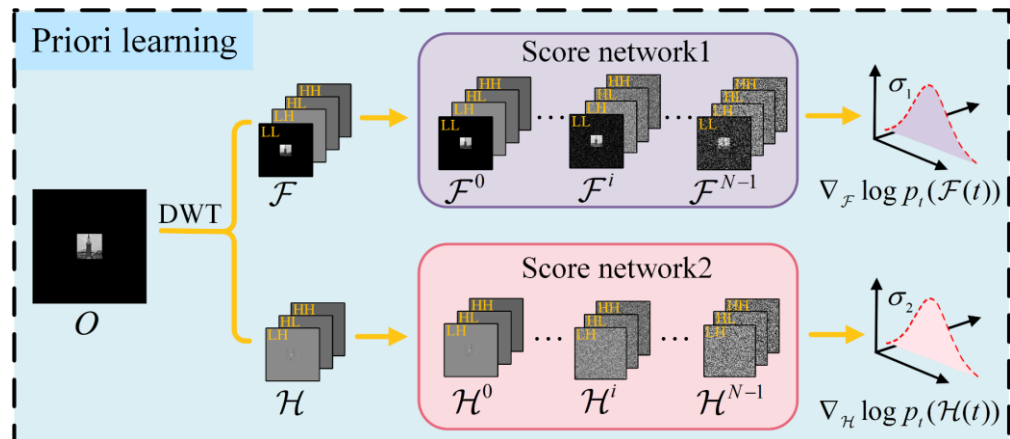


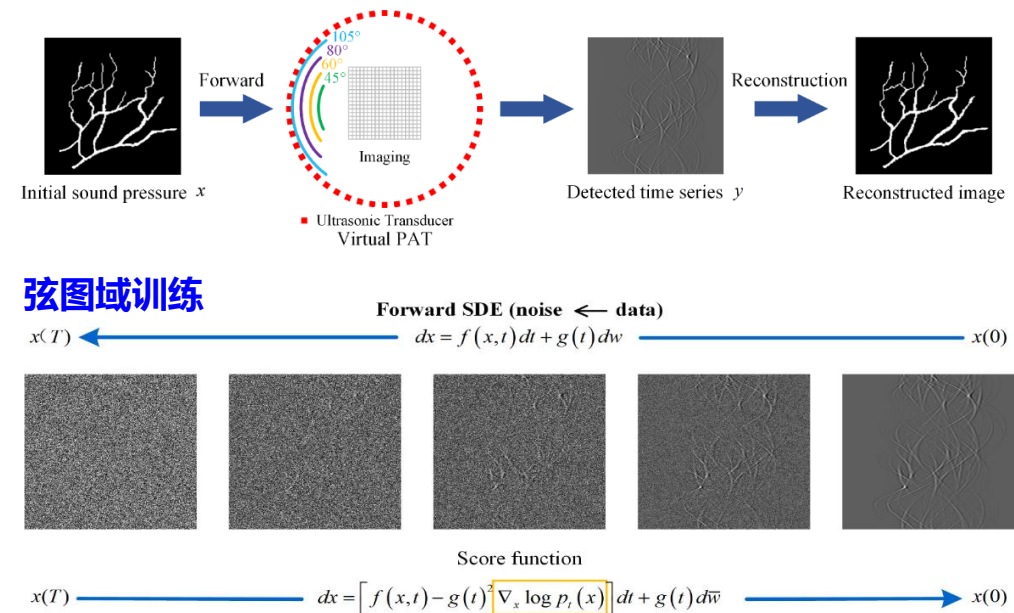
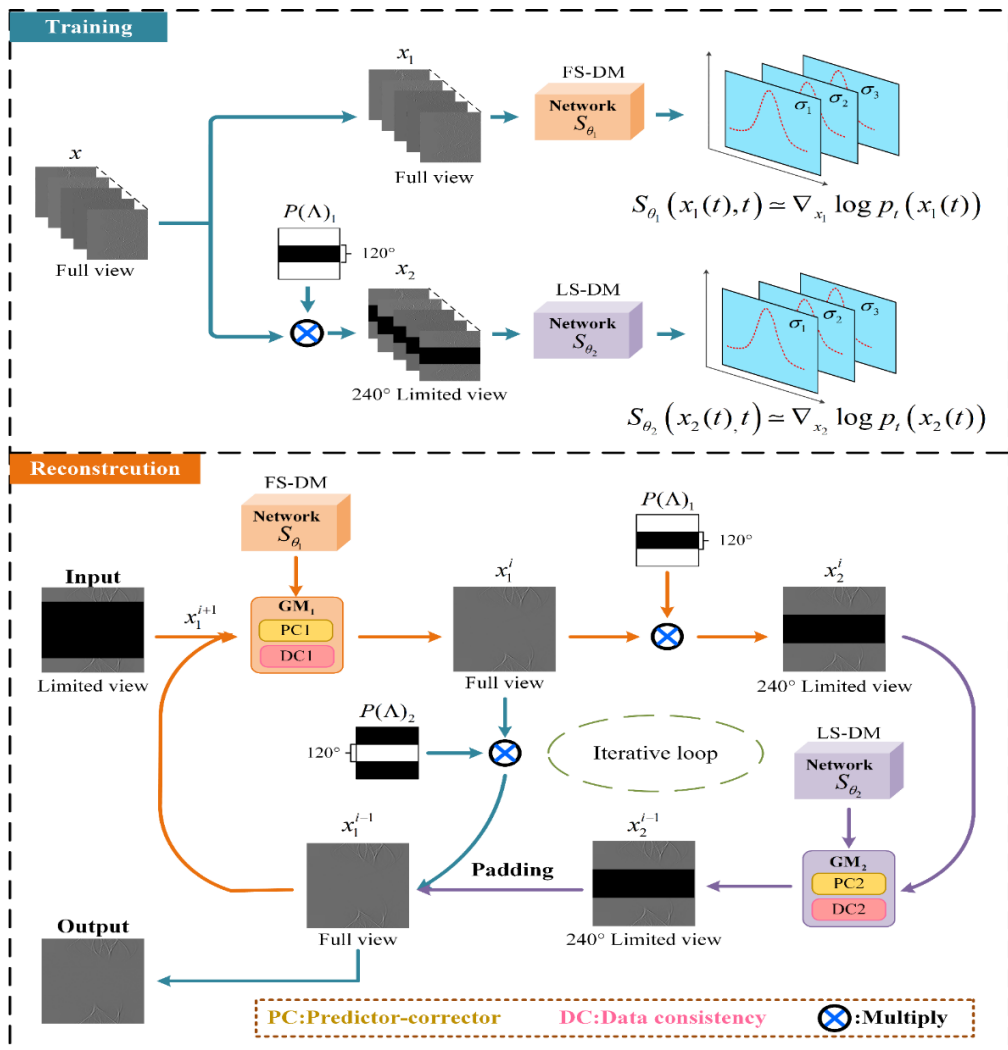
空域先验
信息学习

双域先验信息约束重建过程

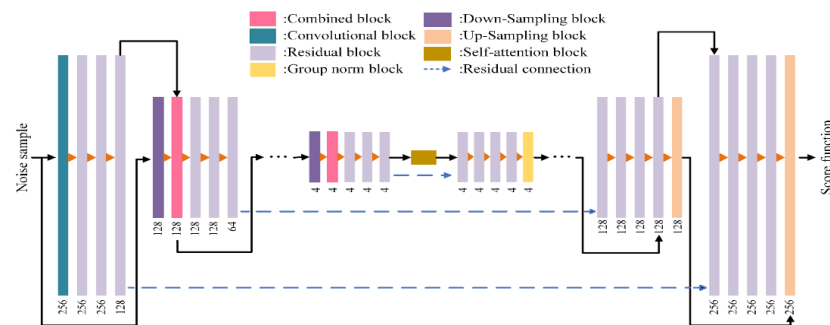
H. Li, et. al., "Dual-domain mean-reverting diffusion model-enhanced temporal compressive coherent diffraction imaging", Optics Express , 2024, 32(9): 15243-15257



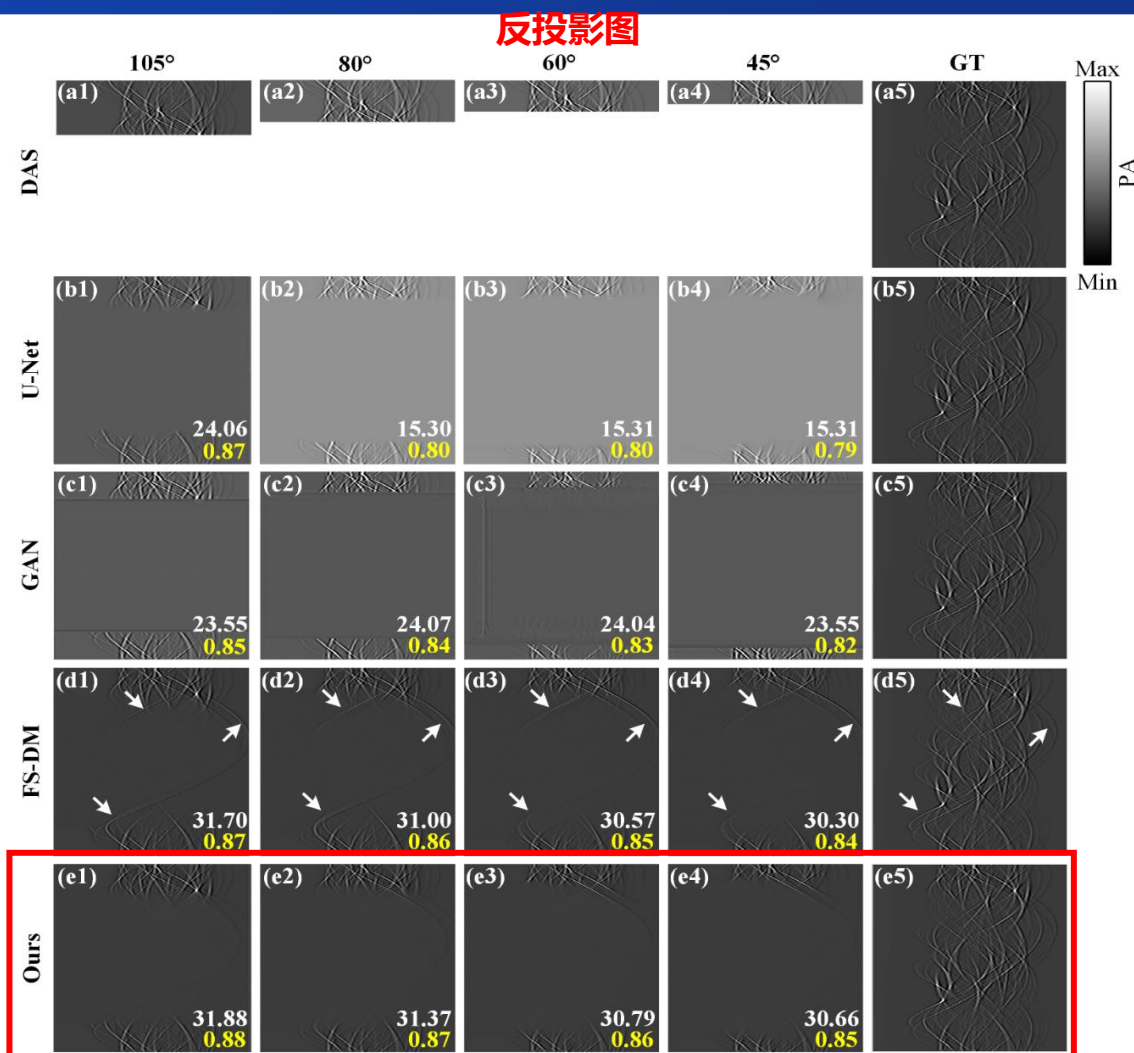
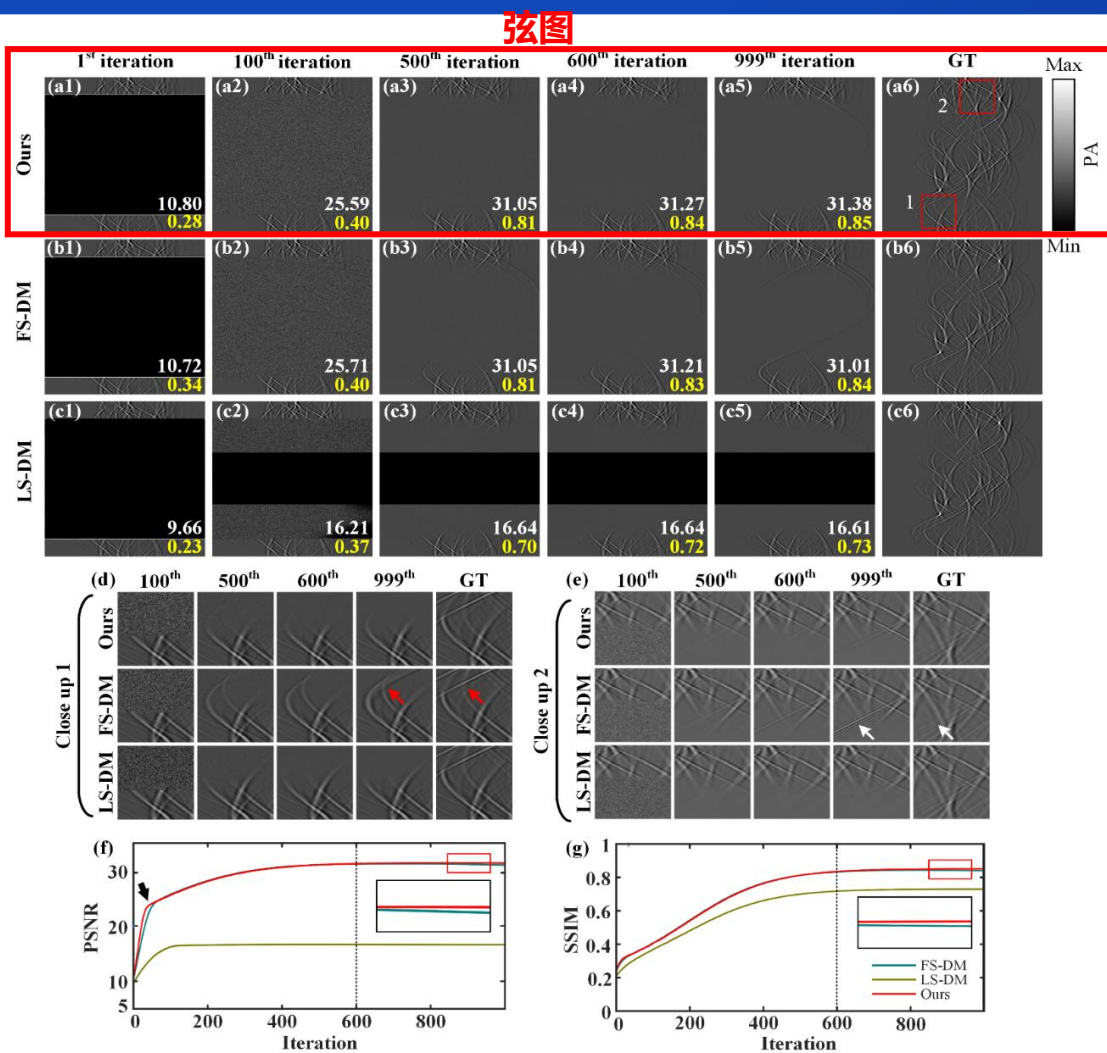




多扩散模型+轮换迭代重建



X. Song, X. Zou, K. Zeng, J. Li, S. Hou, Y. Wu, Z. Li, C. Ma, Z. Zheng, K. Guo, Q. Liu. Multiple diffusion models-enhanced extremely limited-view reconstruction strategy for photoacoustic tomography boosted by multi-scale priors. Photoacoustics, 2024.



X. Song, X. Zou, K. Zeng, J. Li, S. Hou, Y. Wu, Z. Li, C. Ma, Z. Zheng, K. Guo, Q. Liu. Multiple diffusion models-enhanced extremely limited-view reconstruction strategy for photoacoustic tomography boosted by multi-scale priors. Photoacoustics, 2024.



南昌大学成像与视觉表示实验室

Laboratory of Imaging and Vision rEpresentation, LIVE



研究领域：智能成像系统与视觉显示的系统开发与算法研究

- 立足国家大健康战略需求和江西省电子信息产业发展
- 聚焦光电成像系统、软硬件嵌入式开发、信号及视觉处理等行业
- 围绕“传感成像-信号处理-增强显示”中的关键问题与技术瓶颈
- 实现关键技术突破与成果转化



刘且根 教授
国家优青
执行院长
政府津贴获得者



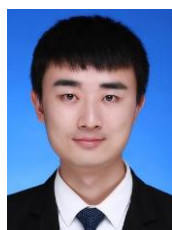
张明辉



徐晓玲



宋贤林



万文博



单文哲



胡嘉文



施柳



肖书源



刘婷婷



吴天涯



王少宇



邹响红

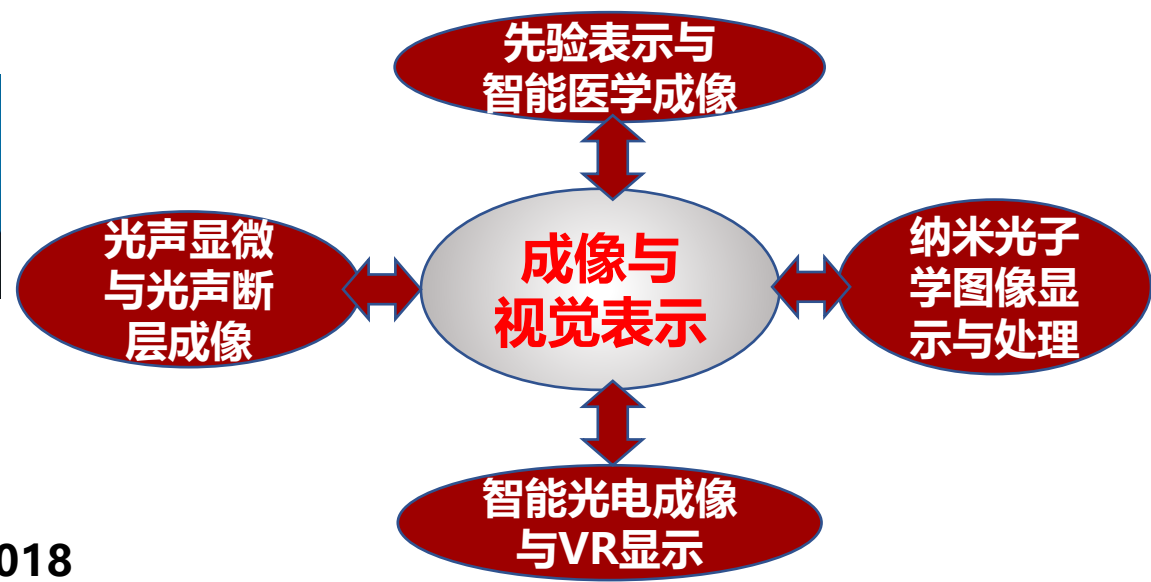
实验室平台及网站：

江西省先进信号处理与智能通信重点实验室

江西省智能医学信息检测与物联工程研究中心

<https://github.com/yqx7150>; <https://www.labxing.com/lab/1018>

- 2024年吴文俊人工智能技术发明一等奖
- 2022年江西省自然科学一等奖
- 2023年江西省青年科学技术奖
- 2022年中国体视学学会青年科技奖
- 2020年中国体视学学会科技进步二等奖
- 第十届吴文俊人工智能科技进步二等奖
- 江西省省级示范研究生导师创新团队



Overview Repositories 129 Projects

Qiegen Liu
yqx7150

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My current research interest is sparse representation, deep learning and their applications in image processing, computer vision and medical imaging.

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Pinned

- Diffusion-Models-for-Medical-Imaging** (Public)
Diffusion Models for Medical Imaging
★ 43 🍴 3
- FSPI-DM** (Public)
High-resolution iterative reconstruction at extremely low sampling rate for Fourier single-pixel imaging via diffusion model
Jupyter Notebook ★ 16 🍴 1
- MLDM** (Public)
Multi-phase FZA Lensless Imaging via Diffusion Model
Python ★ 8 🍴 1
- Intelligent-3D-holography** (Public)
Python ★ 9 🍴 3
- ISDM** (Public)
Imaging through scattering media via generative diffusion model
Python ★ 8 🍴 4
- HoloDiffusion** (Public)
HoloDiffusion's code.
Jupyter Notebook ★ 3 🍴 1

医学成像-光声成像-计算光学成像代表性工作

Diffusion-Models-for-Medical-Imaging (Public)

Code Issues Pull requests Actions Projects Security Insights

Diffusion-Models-for-Photoacoustic-Imaging (Public)

Code Issues Pull requests Actions Projects Security Insights

Diffusion-Models-for-Computational-Optical-Imaging (Public)

main 1 Branch 0 Tags

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Code

qyx7150 README.md d7950ef · 3 days ago 42 Commits

File	Action	Time
figures	Add files via upload	3 days ago
CITA 2024.pdf	Add files via upload	3 days ago
README.md	README.md	3 days ago

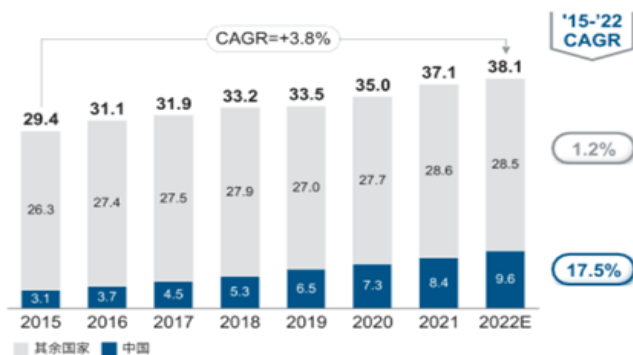
Diffusion-Models-for-Computational-Optical-Imaging

实验室代码公开超过100项:

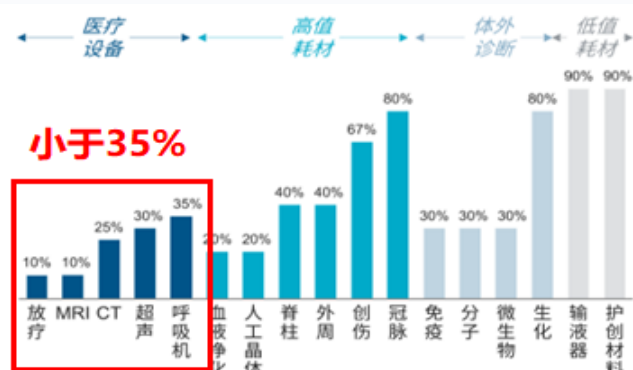
<https://github.com/yqx7150>

研究方向1：先验表示与智能医学成像

大型医疗器械国产化替代需求



全球/中国医疗器械市场规模 (千亿元)

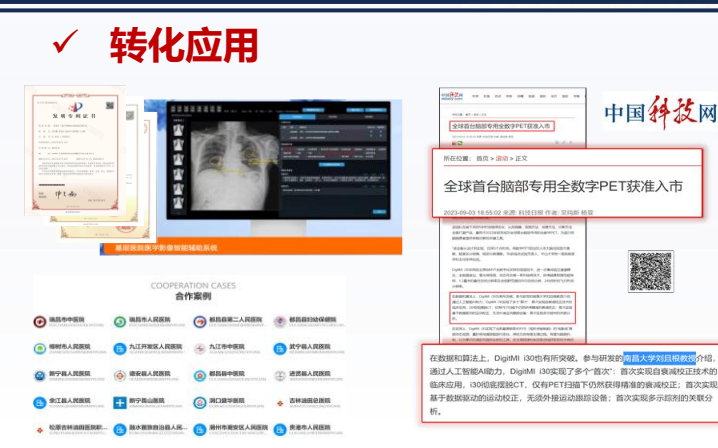
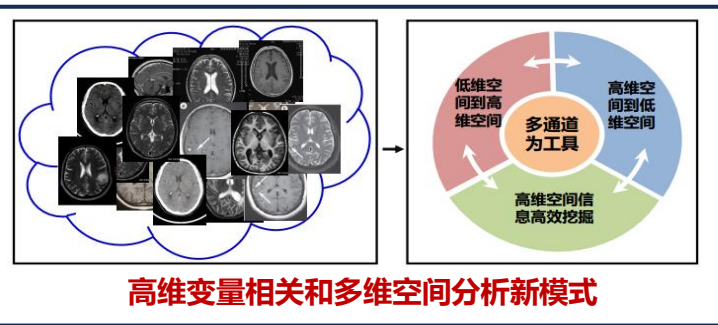


医疗器械细分市场国产品牌占比

“企业-高校-医院”协同创新



- 以高校为主要力量
- 联合企业力量为补充
- 结合医院提出目标任务
- 解决产业链痛点、难点、堵点



研究方向2：光声显微与光声层析成像

◆ 乳腺癌检测主要难点

传统单模态成像精度低，仅实现解剖结构检测

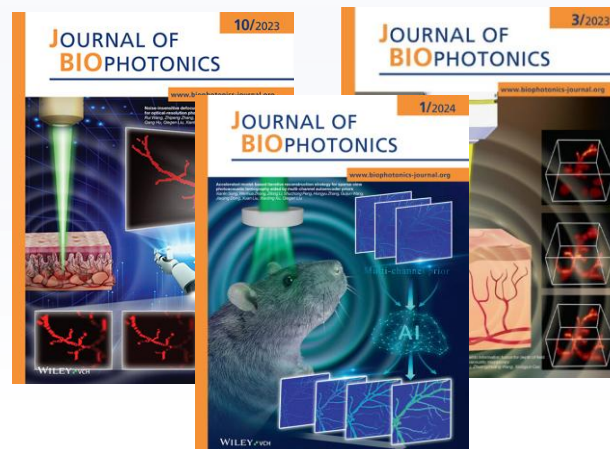
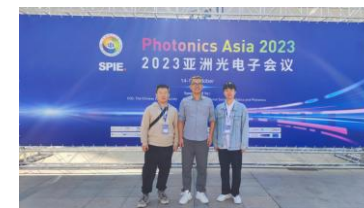
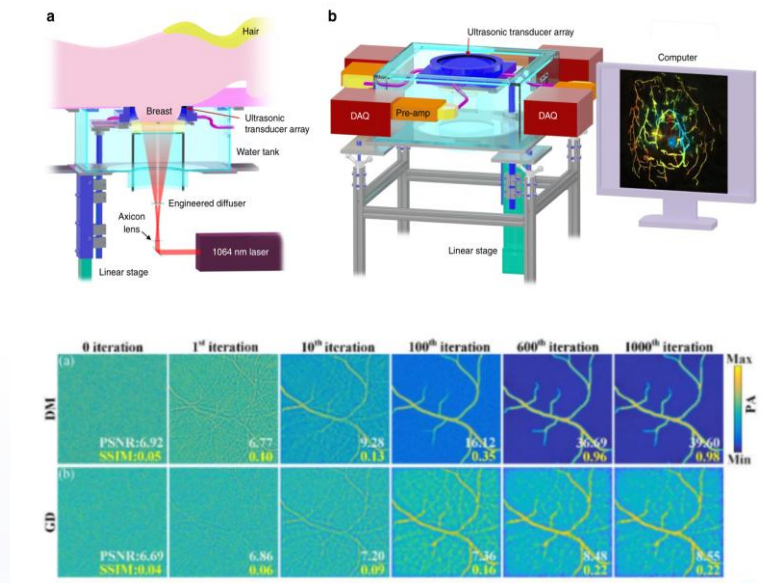
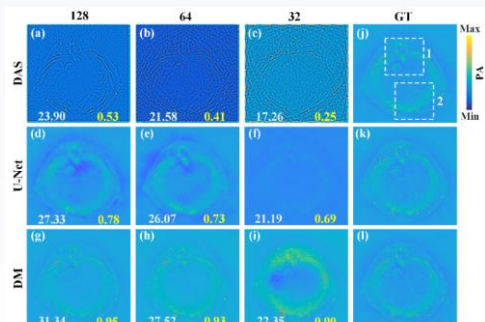
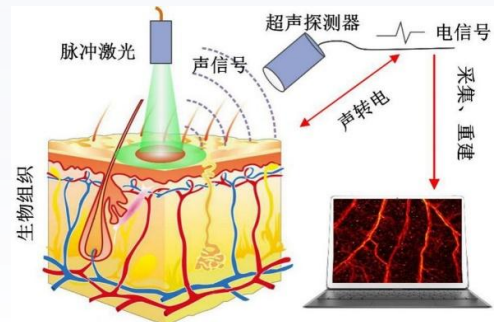
- X光钼靶摄影有电离辐射，超声检测特异性低

◆ 面向乳腺癌早期检测与诊断

- 实现无损、高对比度、高分辨率深层智能成像
- 融合多模态信息，实现特异性诊断

◆ 关键技术难题

- 超声探测器数量有限、受到伪影干扰
- 光学参数背景先验信息缺失



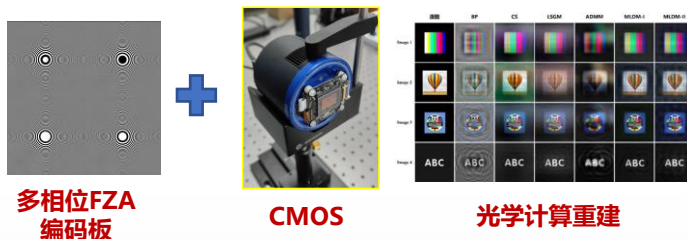
研究方向3：智能光电成像与VR显示

产业需求牵引



复杂场景智能光电成像

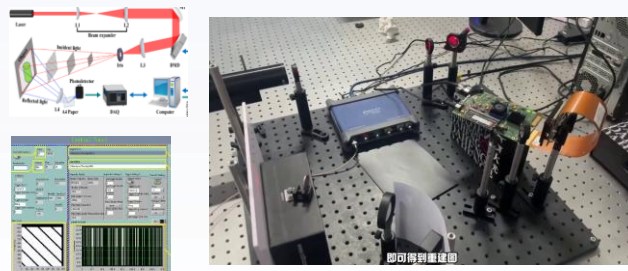
多相位菲涅尔孔径编码无透镜成像



直接飞行时间激光雷达模组

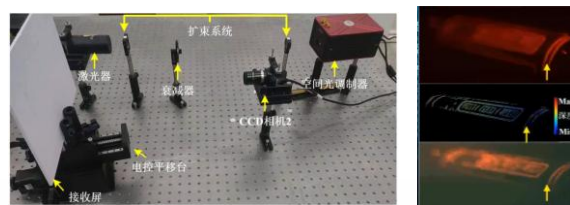


高速追踪单像素探测系统



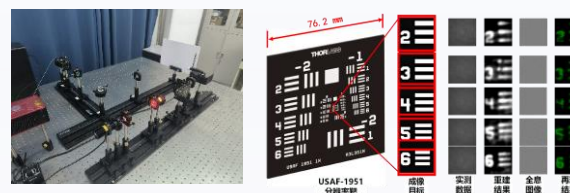
全息VR显示

三维场景快速采集与全息显示



- 极简系统结构
- 超快三维场景采集速度：~ 0.6 s

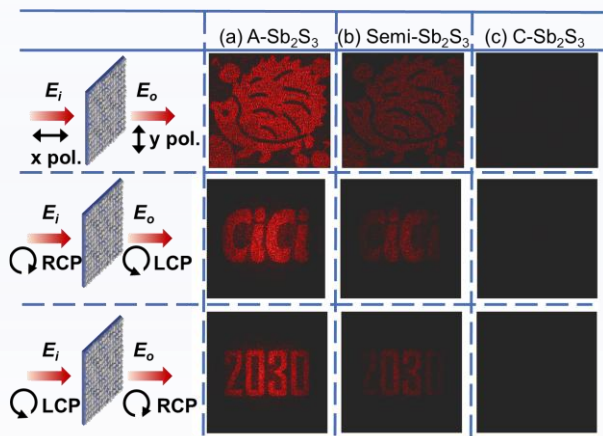
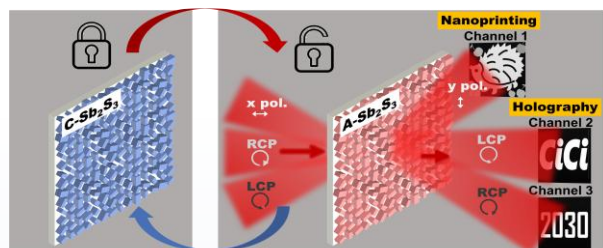
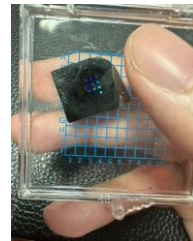
透过散射介质成像与全息显示



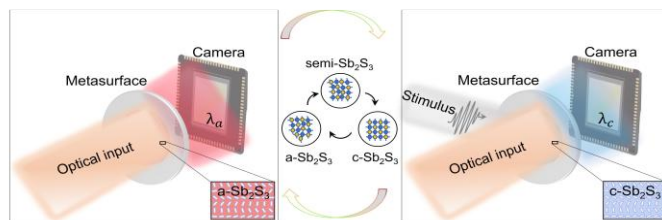
- 透过散射介质实现高分辨率成像
- 实现高质量全息显示

◆ 面向新一代信息技术产业的光学平面集成器件

光子取代电子作为信息载体和能源介质

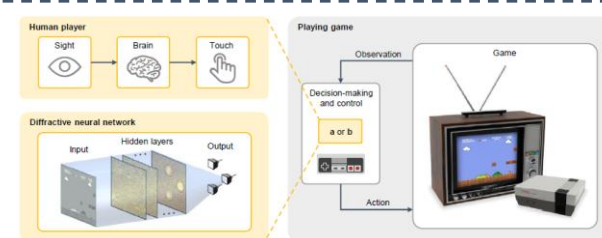


- ✓ 高容量、高集成度的片上信息存储
- ✓ 高空间分辨率的全息图像显示

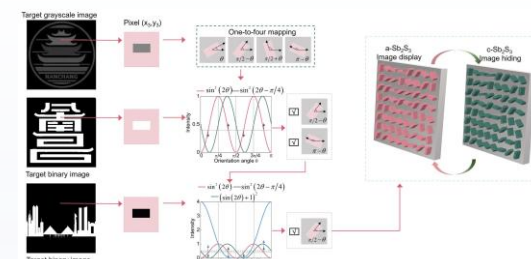


	λ_a	λ_{semi}	λ_c
a-Sb ₂ S ₃			
semi-Sb ₂ S ₃			
c-Sb ₂ S ₃			

- ✓ 光谱和空间调控的高自由度
- ✓ 可调谐波长的全息图像显示



- ✓ 高速度、大带宽、低功耗、高集成
- ✓ 模拟人类的感知能力与控制行为



E_i & E_o	$\leftrightarrow 0 \uparrow \pi/2$	$\leftrightarrow \pi/8 \uparrow 5\pi/8$	$\leftrightarrow 0 \leftrightarrow 0$
a-Sb ₂ S ₃			
c-Sb ₂ S ₃			

- ✓ 信息复用超表面的极简设计
- ✓ 动态信息显示及加密编码



欢迎各位专家指导工作!

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